

Nucleic Acids

Introduction

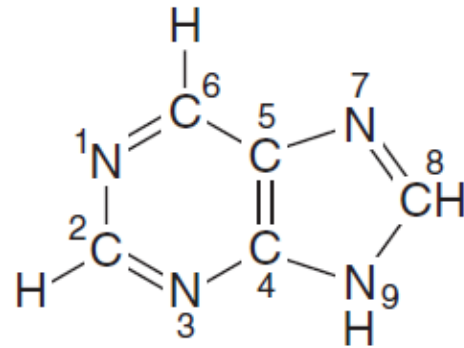
- ❑ A segment of a DNA molecule that contains the information required for the synthesis of a functional biological product, whether protein or RNA, is referred to as a **gene**.
- ❑ A cell typically has many thousands of genes, and DNA molecules, not surprisingly, tend to be very large.
- ❑ The storage and transmission of biological information are the only known functions of DNA

Introduction

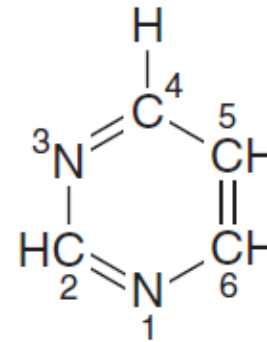
DNA $\xrightarrow{\text{Transcription}}$ RNA $\xrightarrow{\text{Translation}}$ Protein

Nitrogenous bases

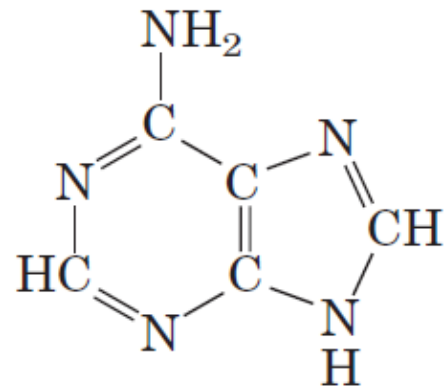
- Purines and pyrimidines are nitrogen-containing **heterocycles**,



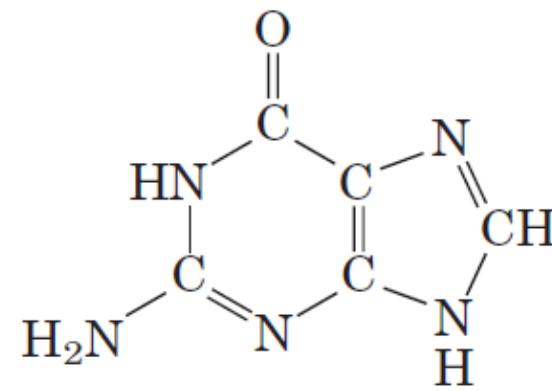
Purine



Pyrimidine

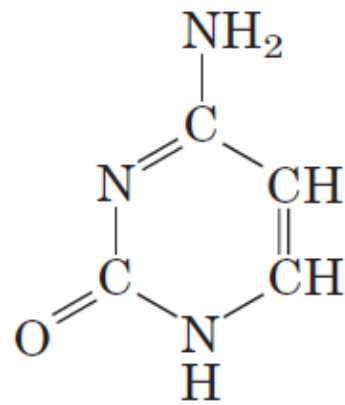


Adenine

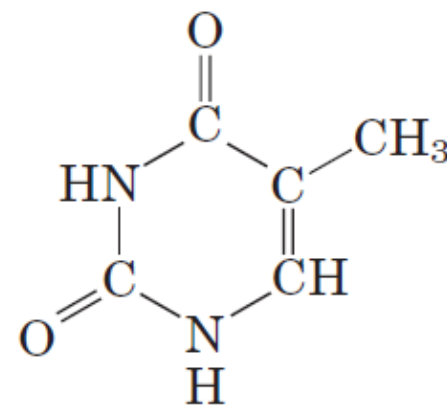


Guanine

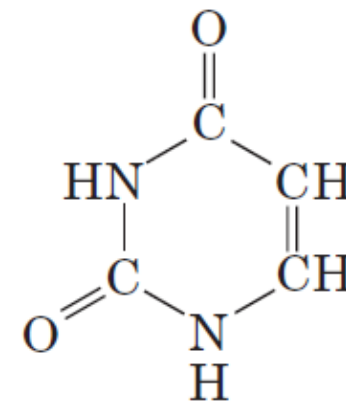
Purines



Cytosine



Thymine
(DNA)



Uracil
(RNA)

Pyrimidines

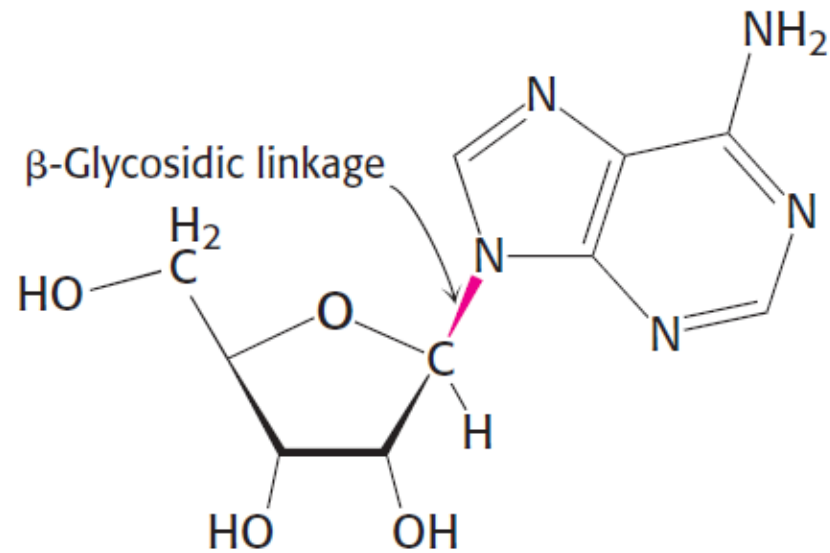
N bases derivatives

- ❑ 5-methylcytosine of bacterial and human DNA, 5-hydroxymethylcytosine of bacterial and viral nucleic acids, and mono- and the di-*N*-methylated adenine and guanine of mammalian messenger RNAs that function in oligonucleotide recognition and in regulating the half-lives of RNAs.
- ❑ Free heterocyclic bases include hypoxanthine, xanthine, and uric acid , intermediates in the catabolism of adenine and guanine.
- ❑ Methylated heterocycles of plants include the xanthine derivatives caffeine of coffee, theophylline of tea, and theobromine of cocoa

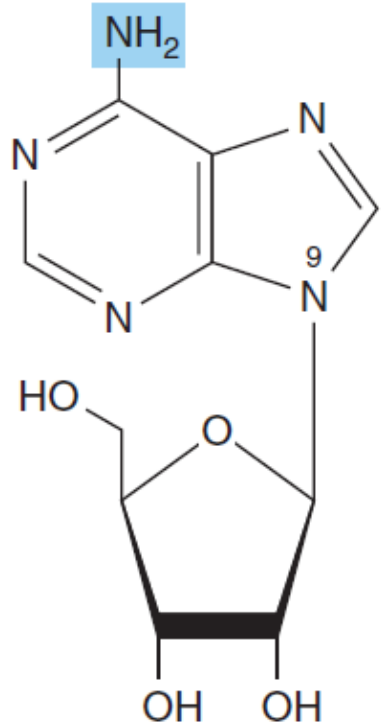
Nucleosides

- ❑ Nucleosides are derivatives of purines and pyrimidines that have a sugar linked to a ring nitrogen of a purine or pyrimidine.
- ❑ Numerals with a prime (eg, 2' or 3') distinguish atoms of the sugar from those of the heterocycle.
- ❑ The sugar in **ribonucleosides** is D-ribose, and in **deoxyribonucleosides** is 2-deoxy-D-ribose.
- ❑ Both sugars are linked to the heterocycle by a **N-glycosidic bond**, almost always to the *N*-1 of a pyrimidine or to *N*-9 of a purine

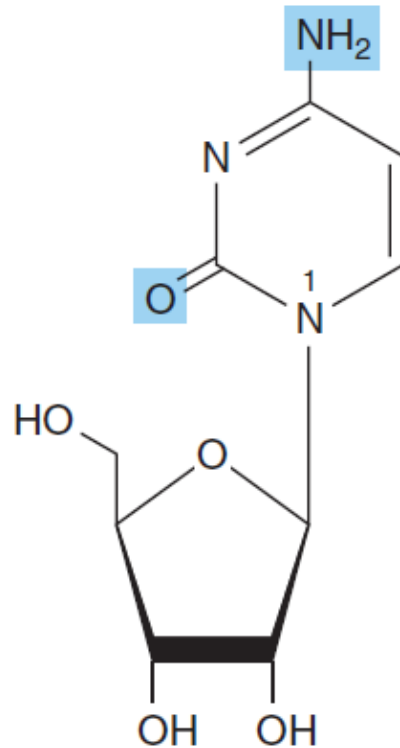
Nucleosides



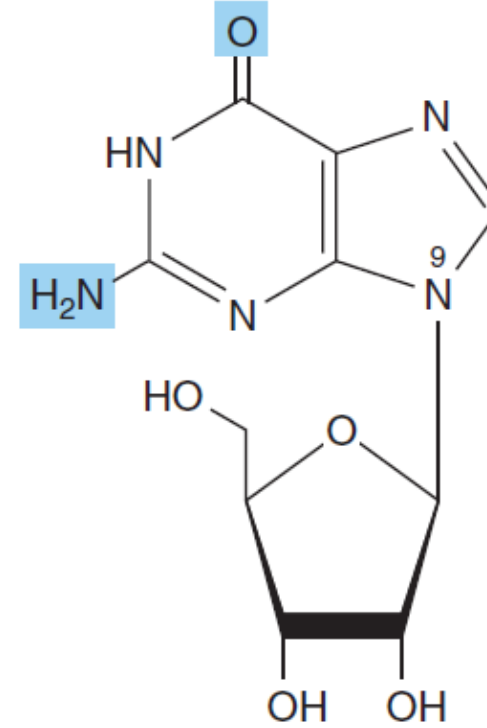
Nucleosides



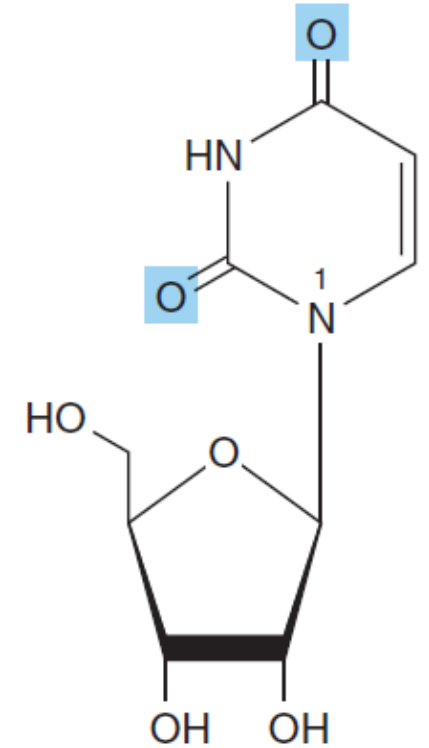
Adenosine



Cytidine



Guanosine



Uridine

Nucleosides

- ❑ **pseudouridine**, a nucleoside in which D-ribose is linked to C-5 of uracil by a **carbon-to-carbon bond** rather than by the usual β -N-glycosidic bond.

Nucleotides

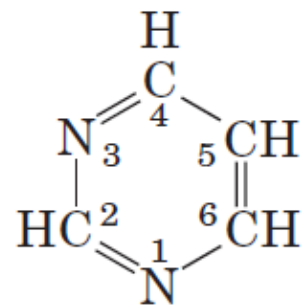
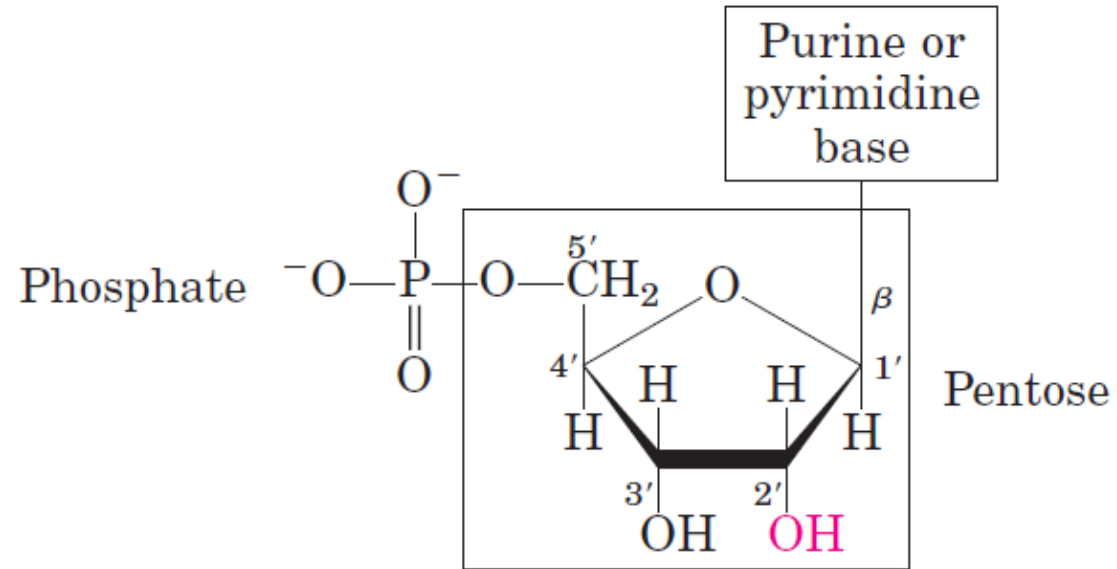
□ have three characteristic components:

(1) a nitrogenous (nitrogen-containing) base

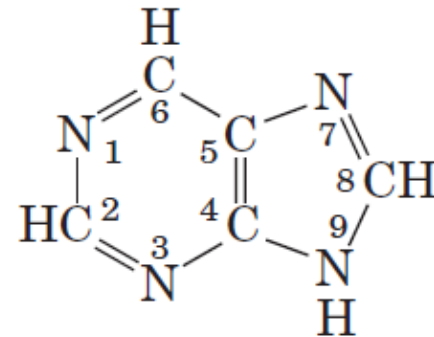
(2) a pentose

(3) a phosphate

Nucleotides



Pyrimidine



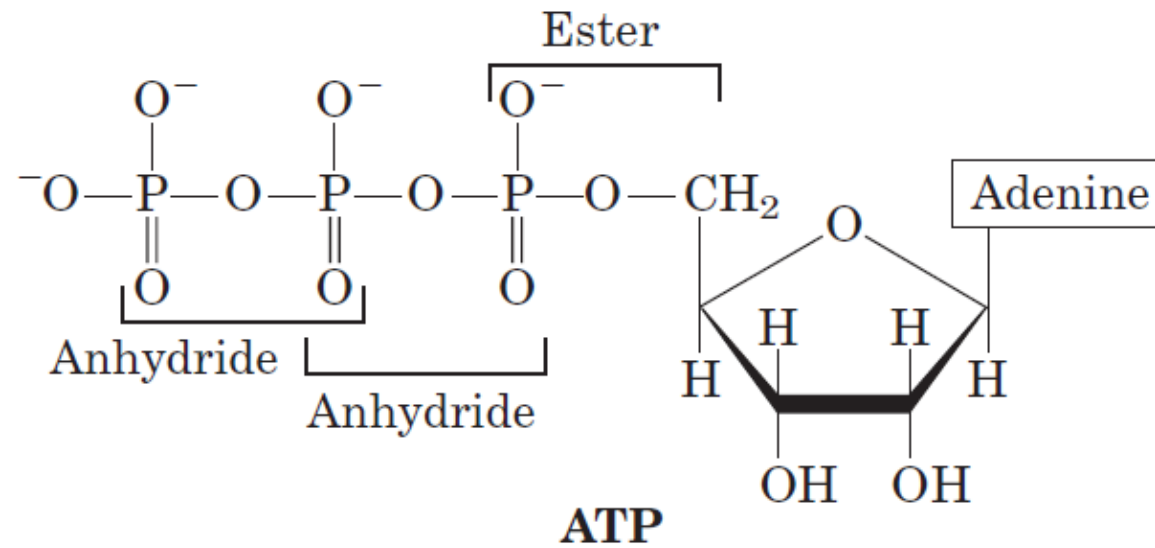
Purine

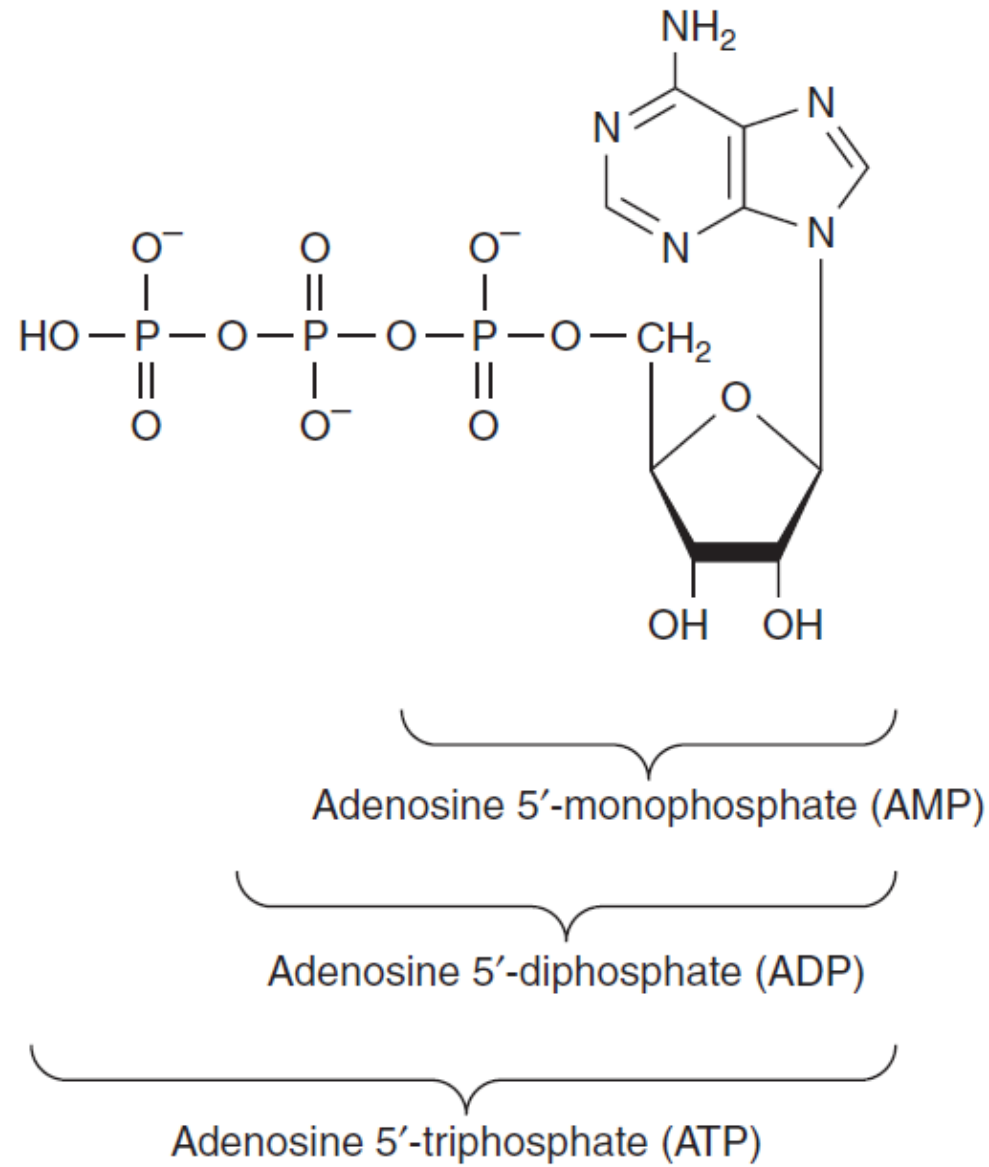
Nucleotides

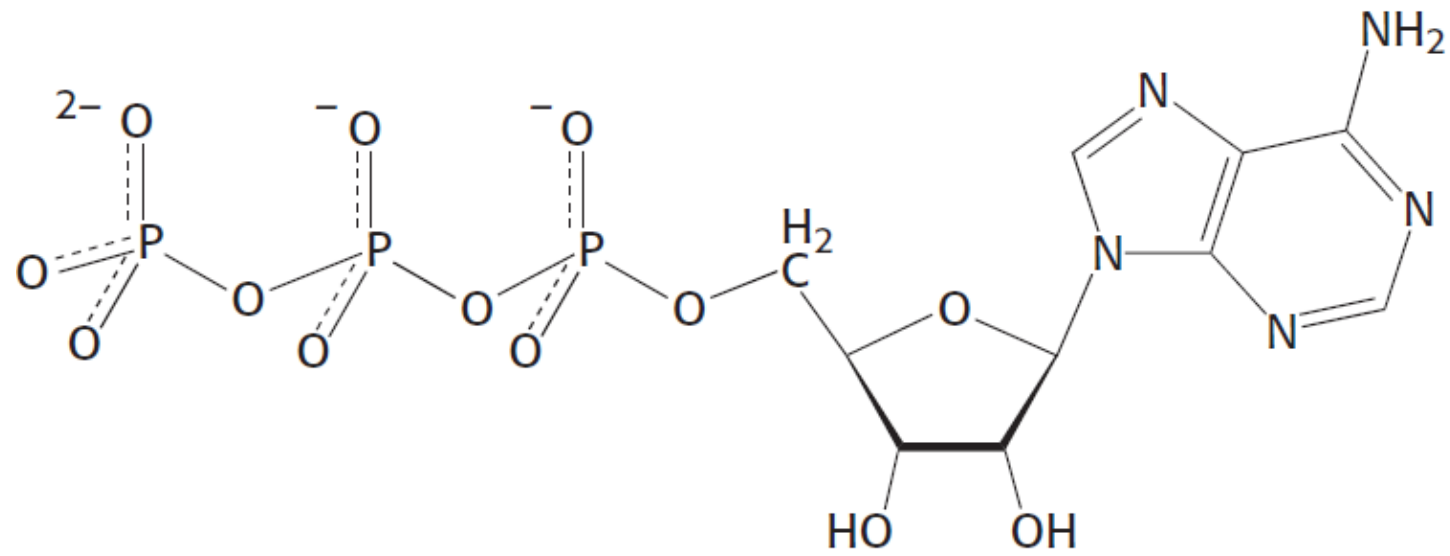
- ❑ The base of a nucleotide is joined covalently (at N-1 of pyrimidines and N-9 of purines) in an *N*—glycosyl bond to the 1 carbon of the pentose, and the phosphate is esterified to the 5 carbon.
- ❑ Pentoses are in their -furanose

Nucleotides

- ❑ Mononucleot**ides** are nucleos**ides** with a phosphoryl group esterified to a hydroxyl group of the sugar. The 3'- and 5'-nucleotides are nucleosides with a phosphoryl group on the 3'- or 5'-hydroxyl group of the sugar, respectively.
- ❑ Most nucleotides are 5'-
- ❑ Additional phosphoryl groups, ligated by **acid anhydride bonds** to the phosphoryl group of a mononucleotide, form **nucleoside diphosphates** and **triphosphates**



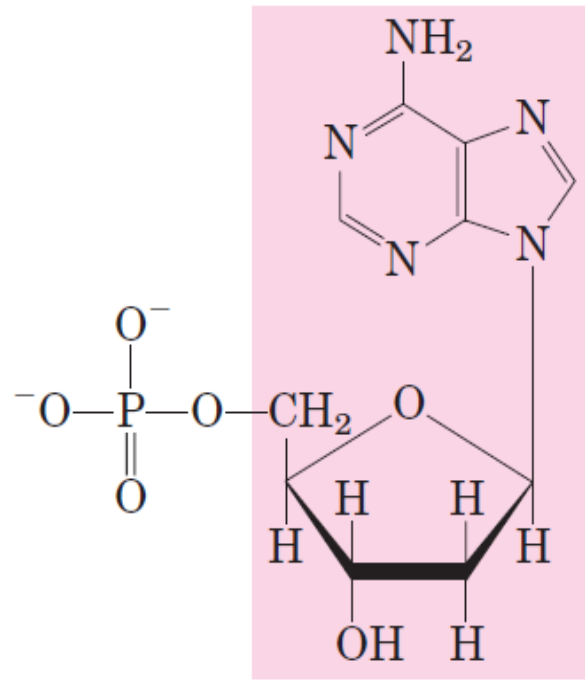




5'-ATP

Nucleotides functions

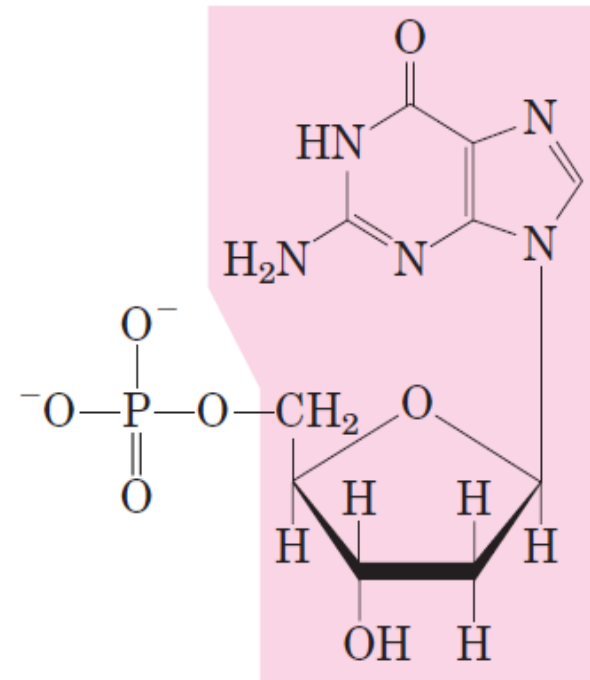
- ❑ Precursors of nucleic acids
- ❑ Derivatives are biosynthetic intermediates
- ❑ Participate in metabolic functions as diverse as energy metabolism, regulation of enzyme activity, form a portion of many coenzymes, the principal donors and acceptors of phosphoryl groups
- ❑ Second messengers and signal transduction
- ❑ Synthetic analogs in the chemotherapy of cancer and AIDS, and as suppressors of the immune response during organ transplantation



Nucleotide: Deoxyadenylate
(deoxyadenosine
5'-monophosphate)

Symbols: A, dA, dAMP

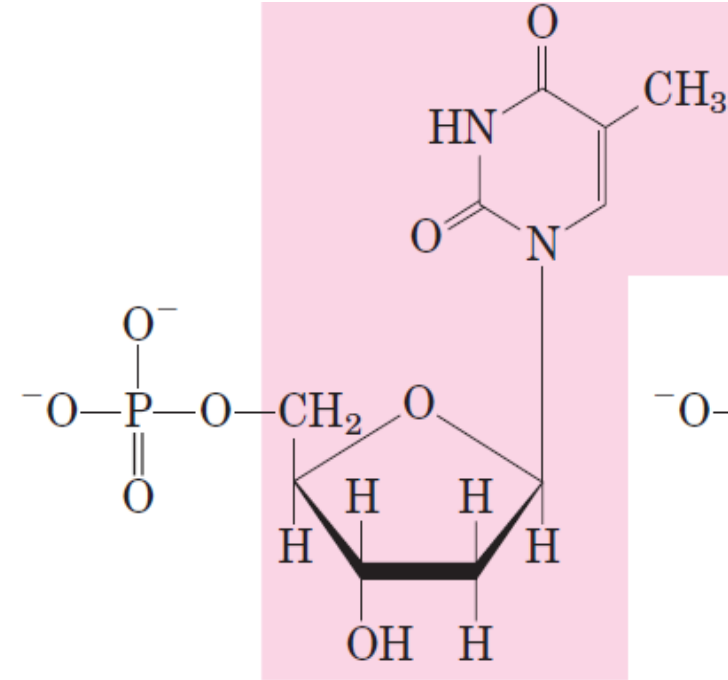
Nucleoside: Deoxyadenosine



Nucleotide: Deoxyguanylate
(deoxyguanosine
5'-monophosphate)

Symbols: G, dG, dGMP

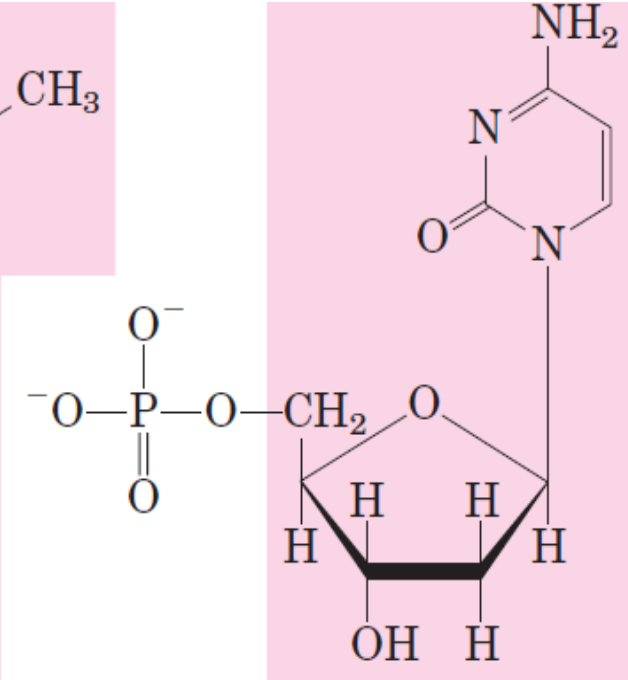
Nucleoside: Deoxyguanosine



Nucleotide: Deoxythymidylate
(deoxythymidine
5'-monophosphate)

Symbols: T, dT, dTMP

Nucleoside: Deoxythymidine

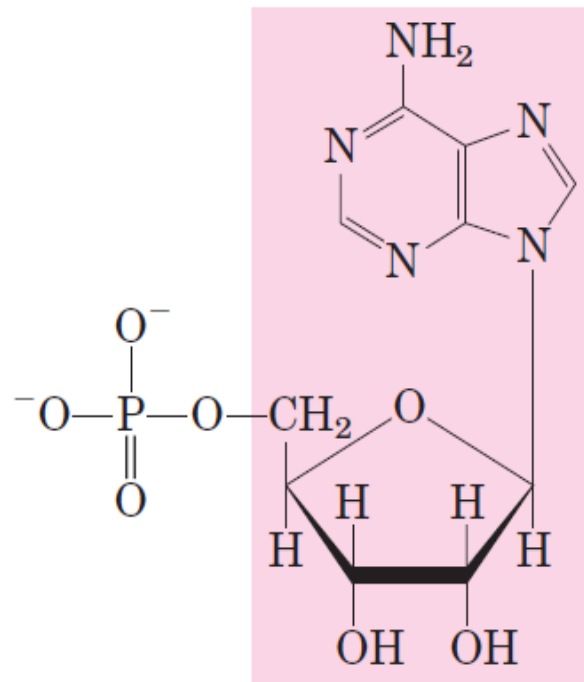


Nucleotide: Deoxycytidylate
(deoxycytidine
5'-monophosphate)

Symbols: C, dC, dCMP

Nucleoside: Deoxycytidine

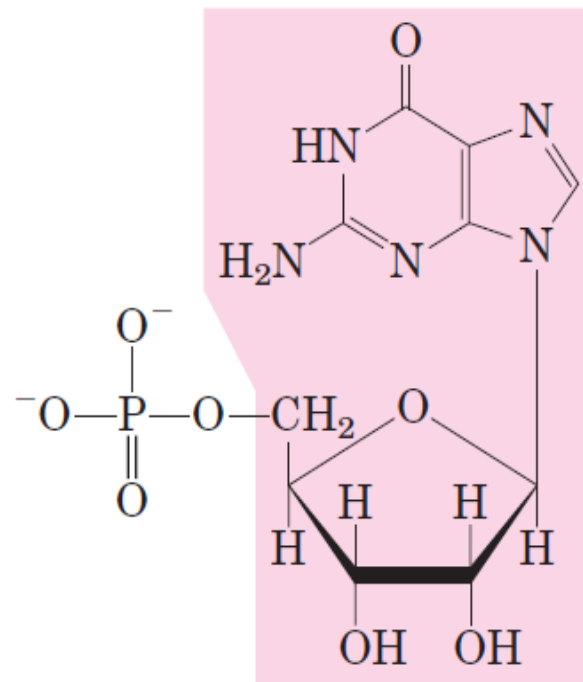
(a) Deoxyribonucleotides



Nucleotide: Adenylate (adenosine 5'-monophosphate)

Symbols: A, AMP

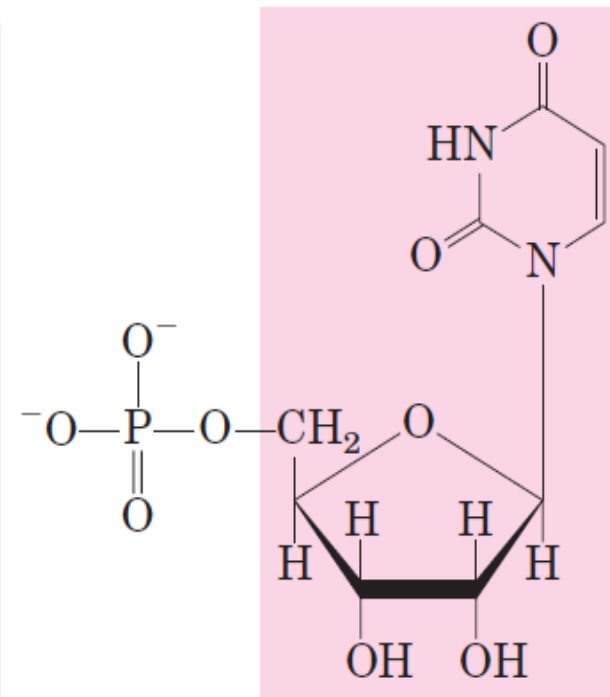
Nucleoside: Adenosine



Nucleotide: Guanylate (guanosine 5'-monophosphate)

Symbols: G, GMP

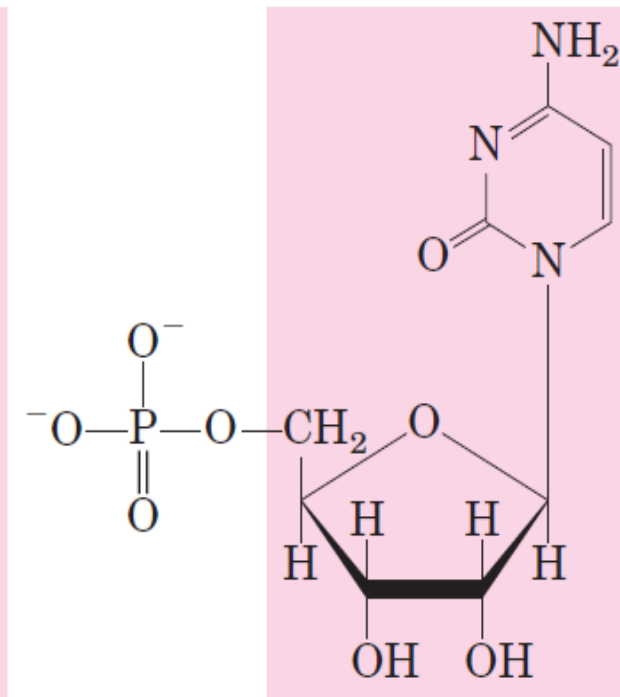
Nucleoside: Guanosine



Nucleotide: Uridylate (uridine 5'-monophosphate)

Symbols: U, UMP

Nucleoside: Uridine

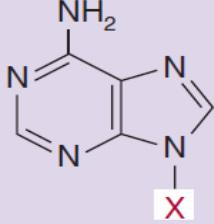
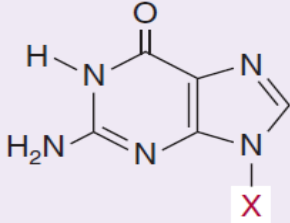
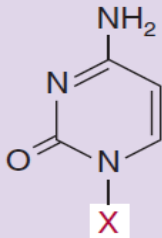
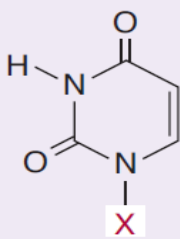
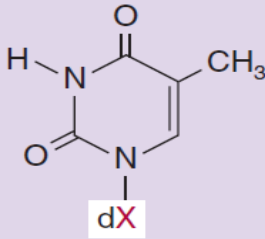


Nucleotide: Cytidylate (cytidine 5'-monophosphate)

Symbols: C, CMP

Nucleoside: Cytidine

(b) Ribonucleotides

Purine or Pyrimidine	X = H	X = Ribose	X = Ribose Phosphate
	Adenine	Adenosine	Adenosine monophosphate (AMP)
	Guanine	Guanosine	Guanosine monophosphate (GMP)
	Cytosine	Cytidine	Cytidine monophosphate (CMP)
	Uracil	Uridine	Uridine monophosphate (UMP)
	Thymine	Thymidine	Thymidine monophosphate (TMP)

SYNTHETIC NUCLEOTIDE ANALOGS

- ▣ inhibition of enzymes essential for nucleic acid synthesis or their incorporation into nucleic acids with resulting disruption of base pairing.
- ▣ Oncologists employ 5-fluoro- or 5-iodouracil, 3-deoxyuridine, 6-thioguanine and 6-mercaptopurine, 5- or 6-azauridine, 5- or 6-azacytidine, and 8-azaguanine), which are incorporated into DNA prior to cell division.
- ▣ The purine analog allopurinol, used in treatment of hyperuricemia and gout.
- ▣ Cytarabine is used in chemotherapy of cancer
- ▣ Azathioprine, which is catabolized to 6-mercaptopurine, is employed during organ transplantation to suppress immunologic rejection

Nucleic acids

- The 5'-phosphoryl group of a mononucleotide can esterify a second hydroxyl group, forming a **phosphodiester**. Most commonly, this second hydroxyl group is the 3'-OH of the pentose of a second nucleotide. This forms a **dinucleotide** in which the pentose moieties are linked by a 3',5'-phosphodiester bond to form the “backbone” of RNA and DNA

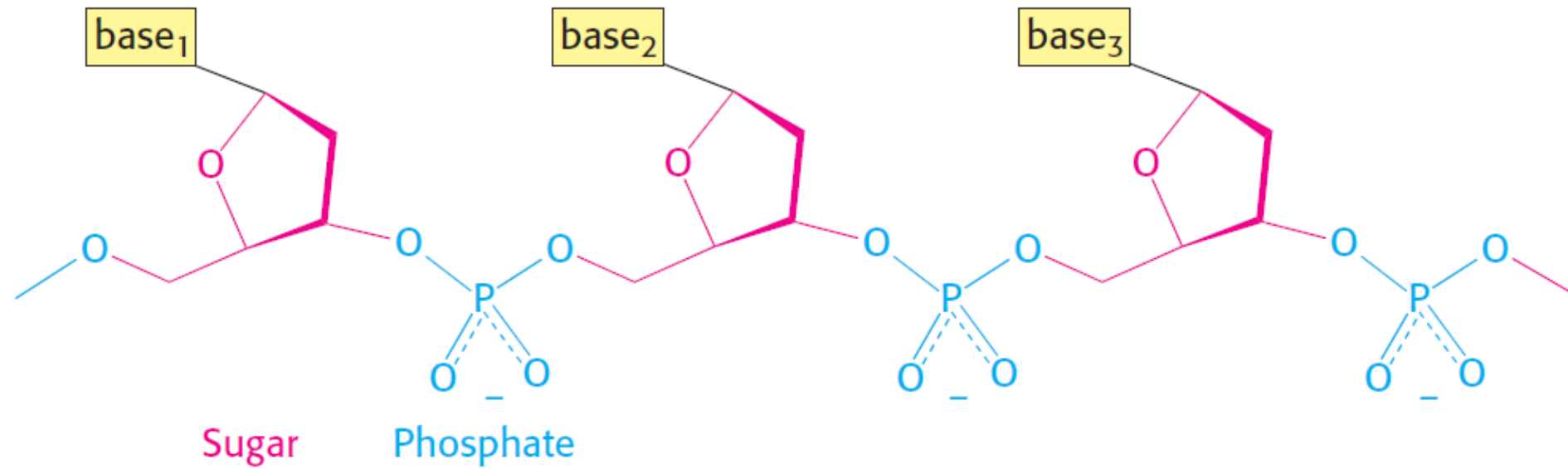
Nucleic acids

- Nucleic acids have two kinds of pentoses.
- 1. The recurring deoxyribonucleotide units of DNA contain 2-deoxy-D-ribose
- 2. the ribonucleotide units of RNA contain D-ribose.

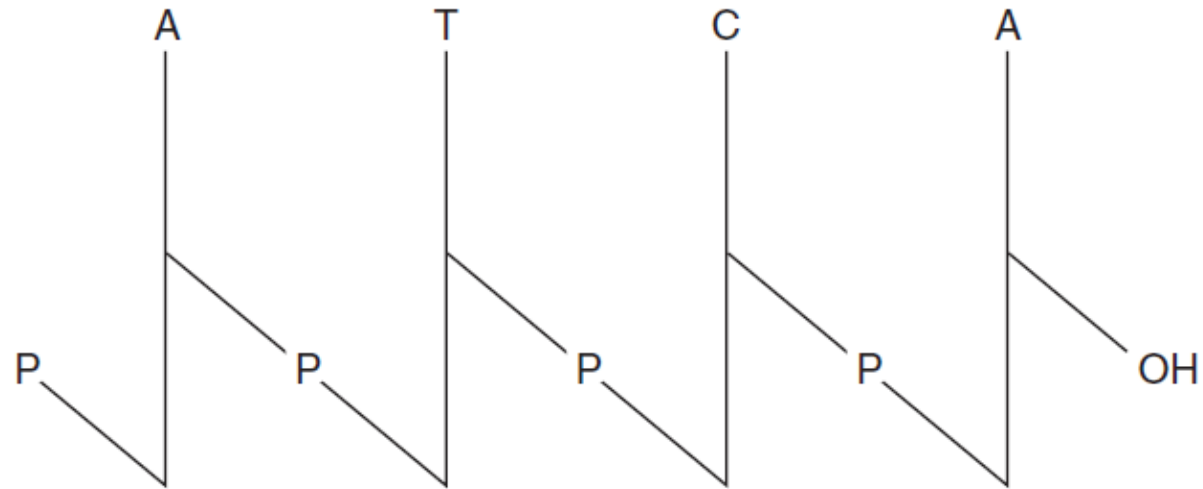
Nucleic acids

- The covalent backbones of nucleic acids consist of alternating phosphate and pentose residues, and the nitrogenous bases may be regarded as side groups joined to the backbone at regular intervals.
- The backbones of both DNA and RNA are hydrophilic. The hydroxyl groups of the sugar residues form hydrogen bonds with water.
- The phosphate groups, with a pK_a near 0, are completely ionized and negatively charged at pH 7, and the negative charges are generally neutralized by ionic interactions with positive charges on proteins, metal ions, and polyamines

Nucleic acids



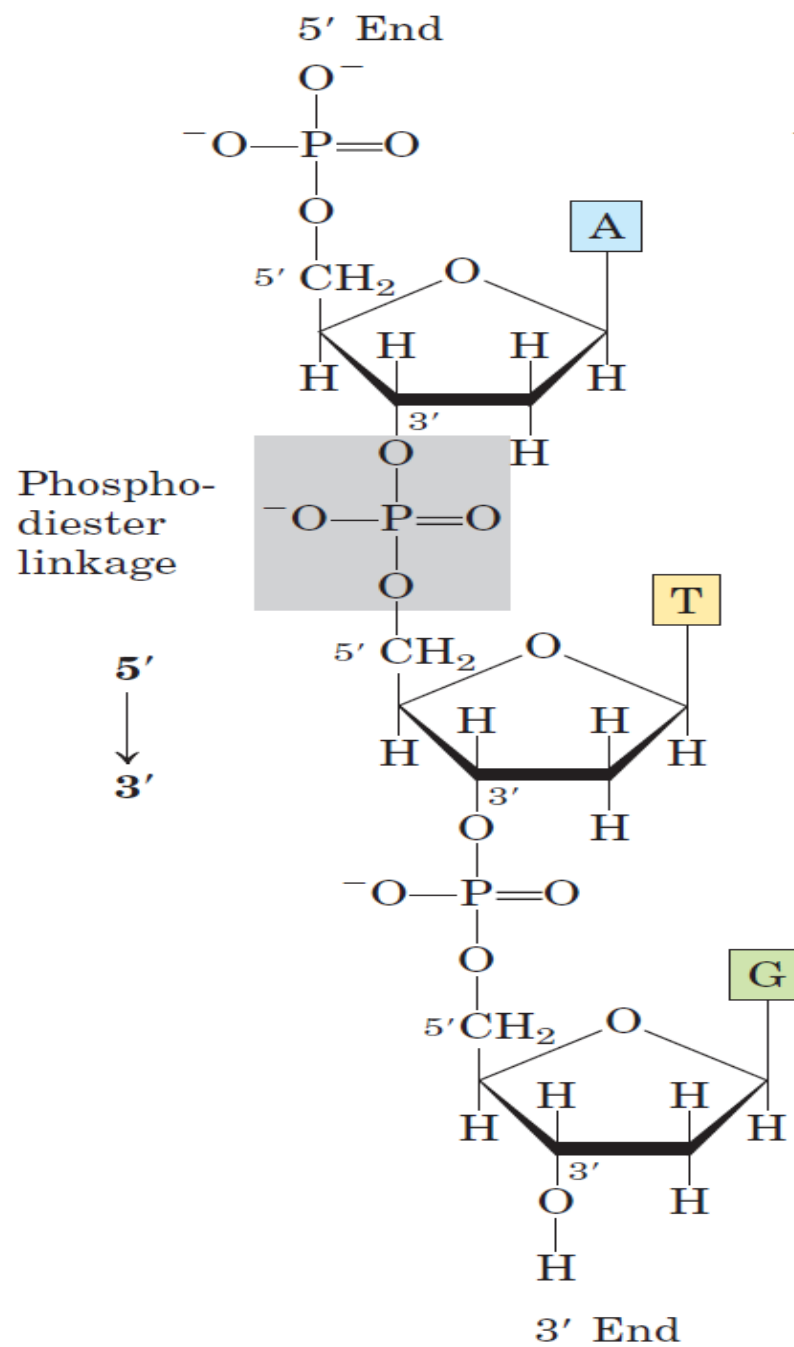
Nucleic acids



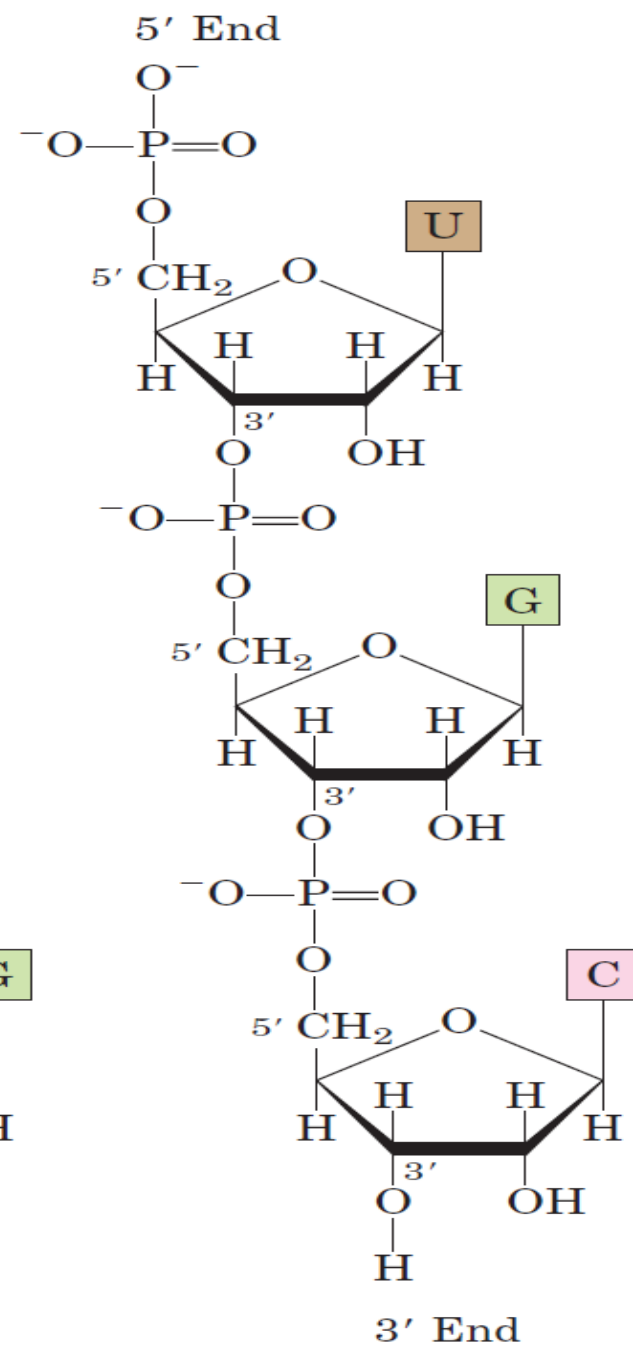
DNA and RNA

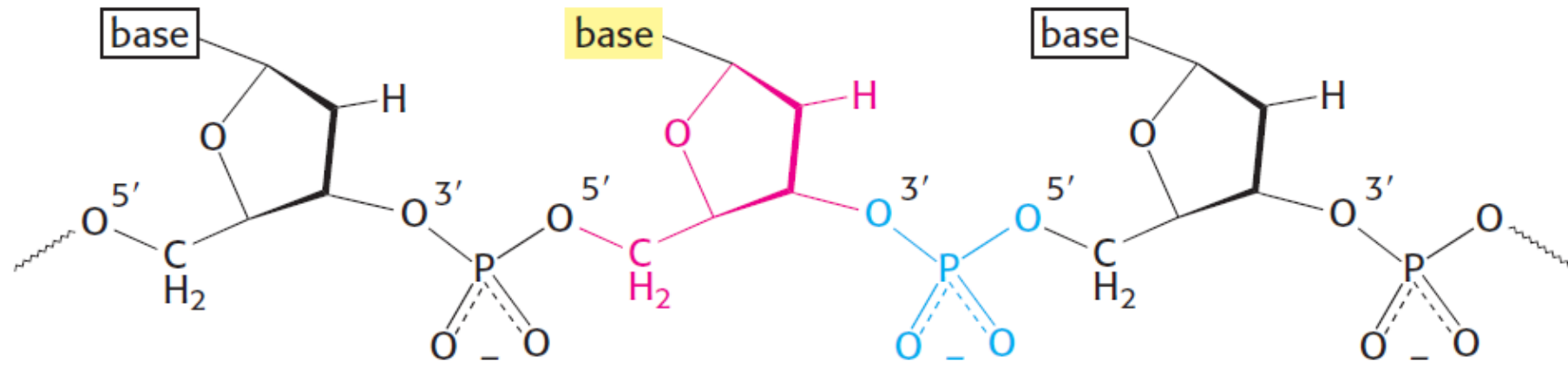
- Both DNA and RNA contain two major purine bases, **adenine** (A) and **guanine** (G), and two major pyrimidines.
- In both DNA and RNA one of the pyrimidines is **cytosine** (C), but the second major pyrimidine is not the same in both: it is **thymine** (T) in DNA and **uracil** (U) in RNA.
- Only rarely does thymine occur in RNA or uracil in DNA

DNA

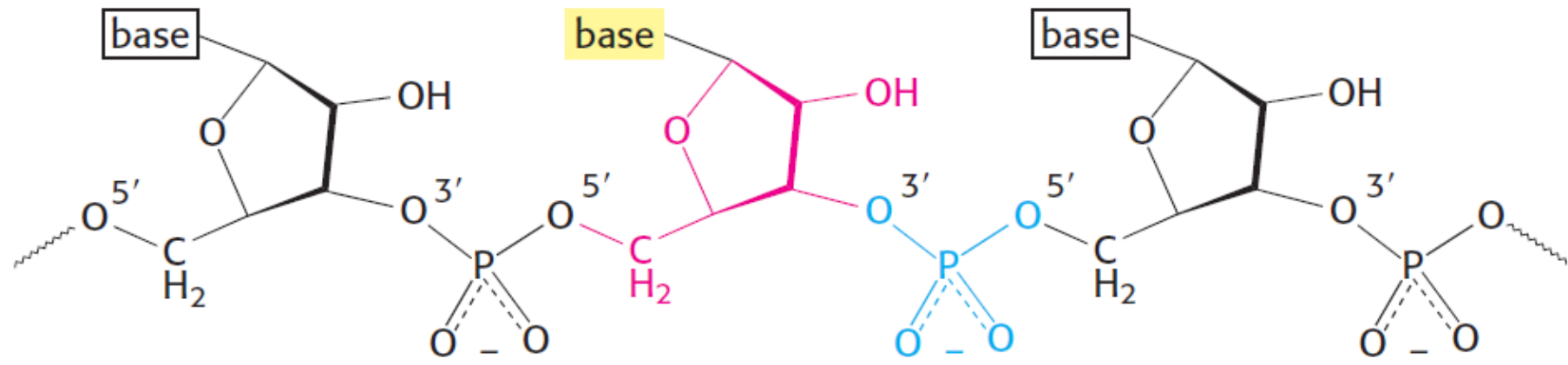


RNA





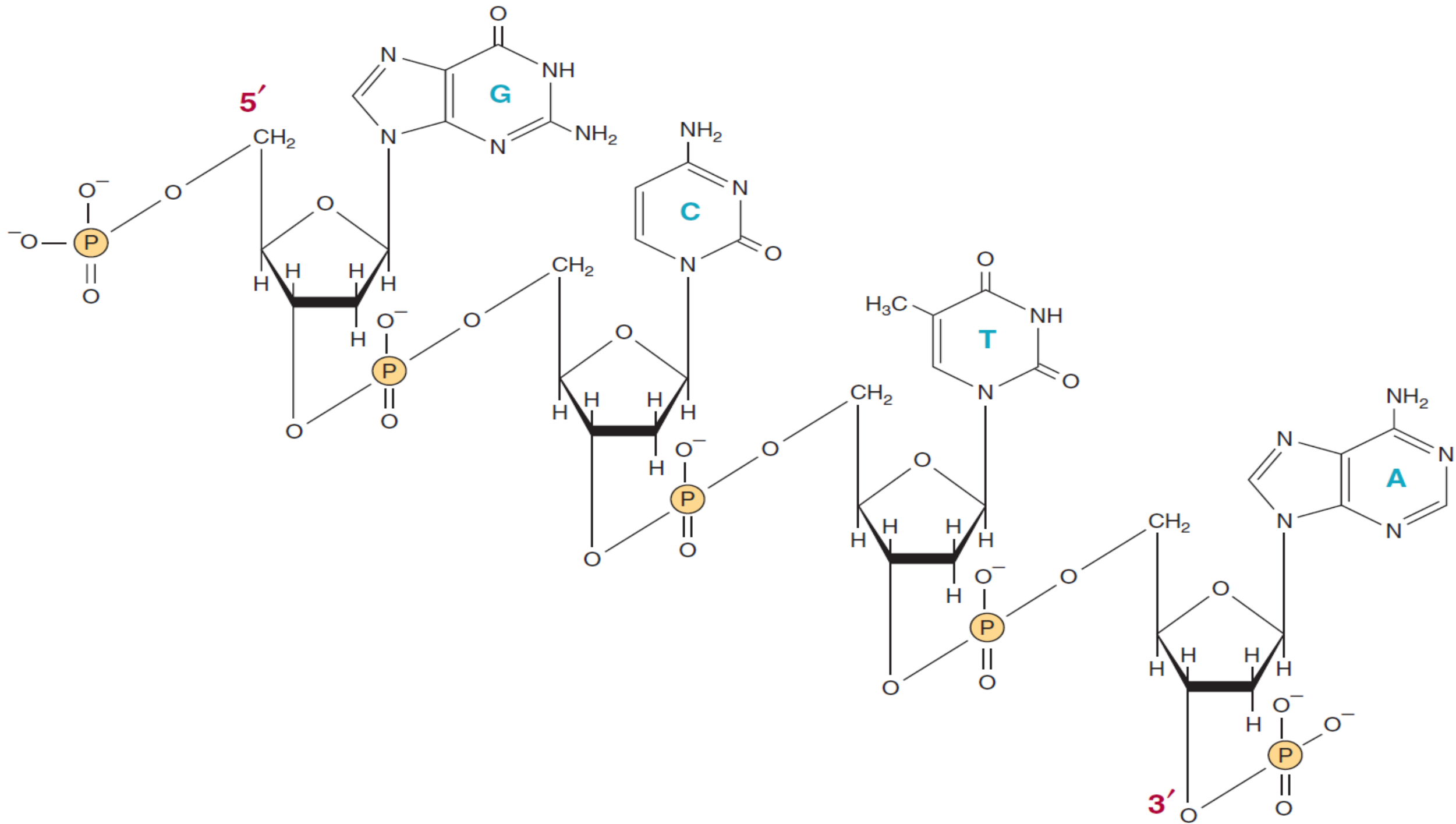
DNA

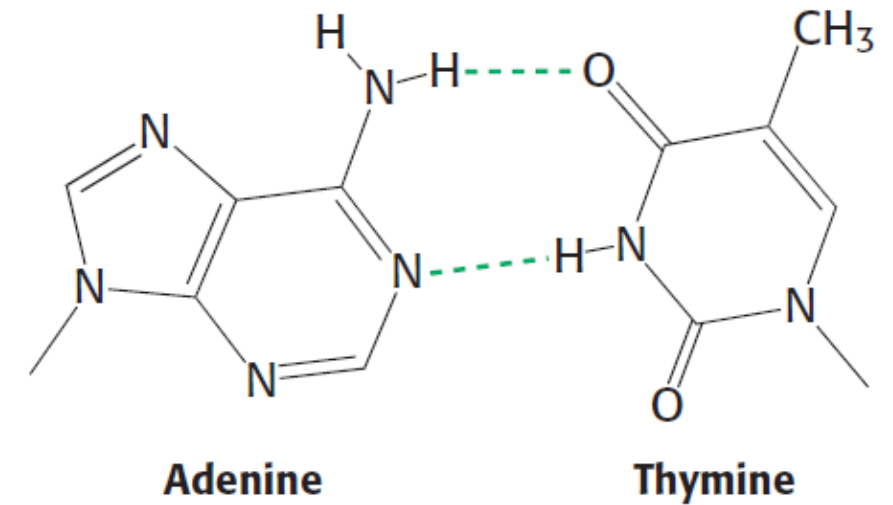
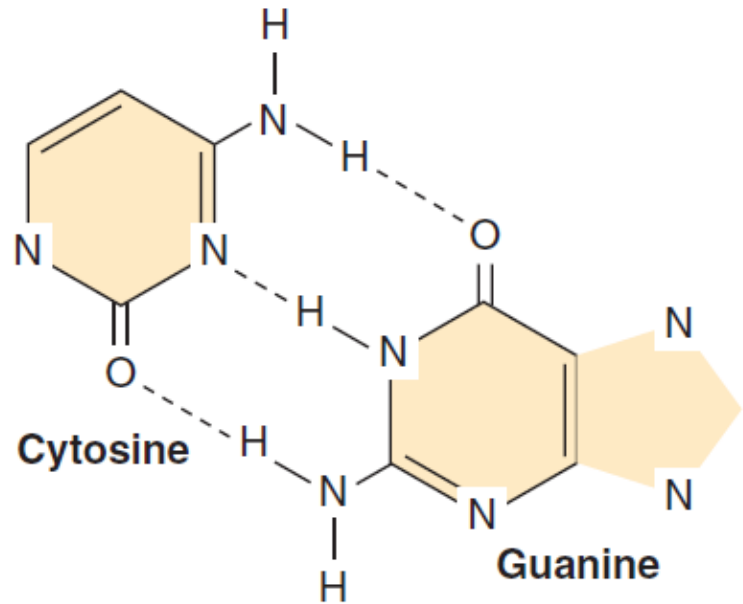
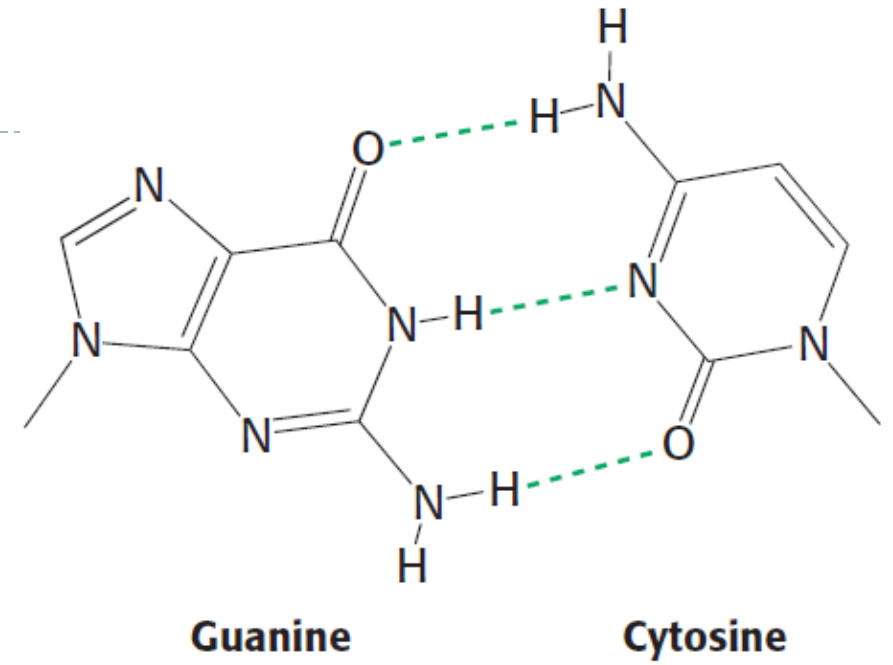
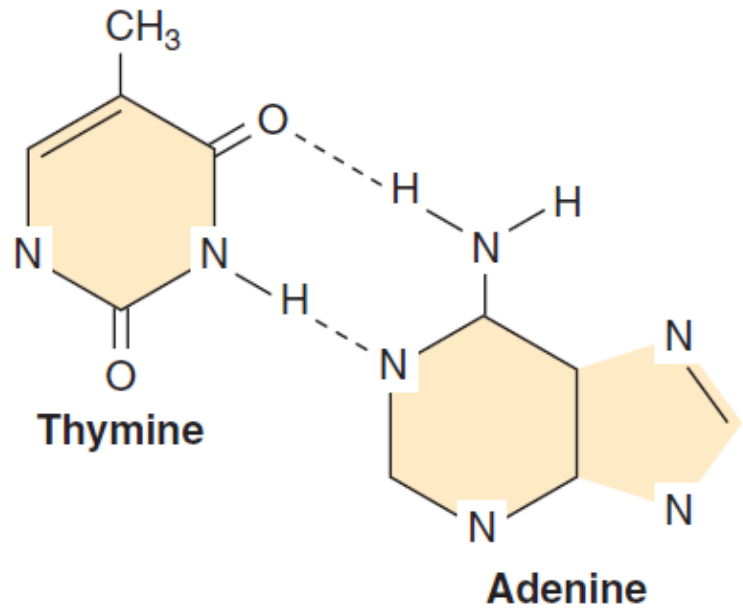


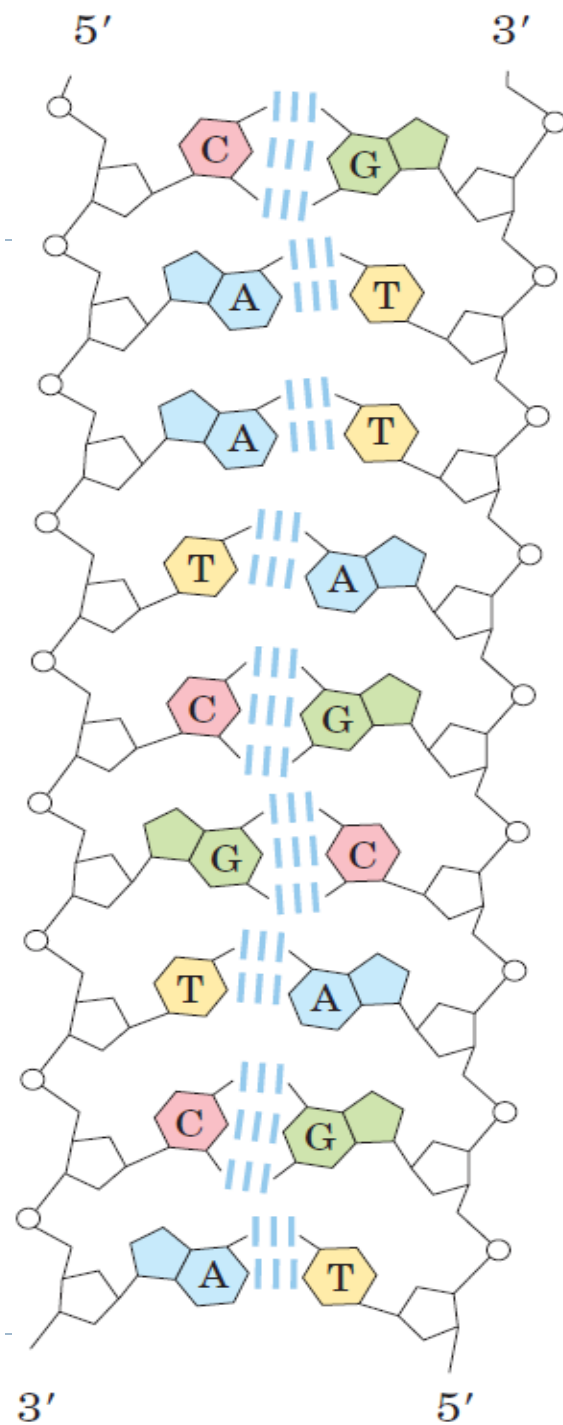
RNA

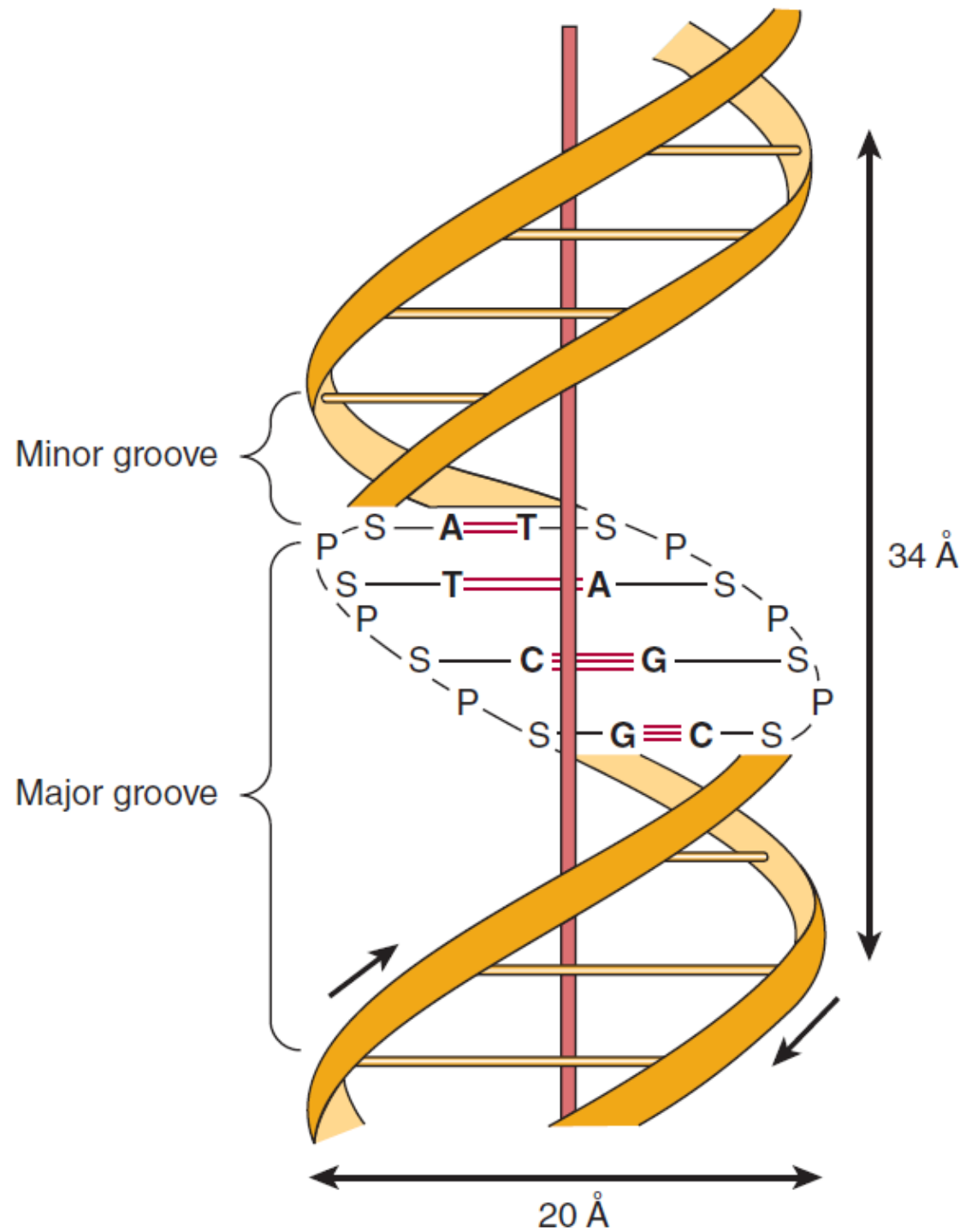
DNA

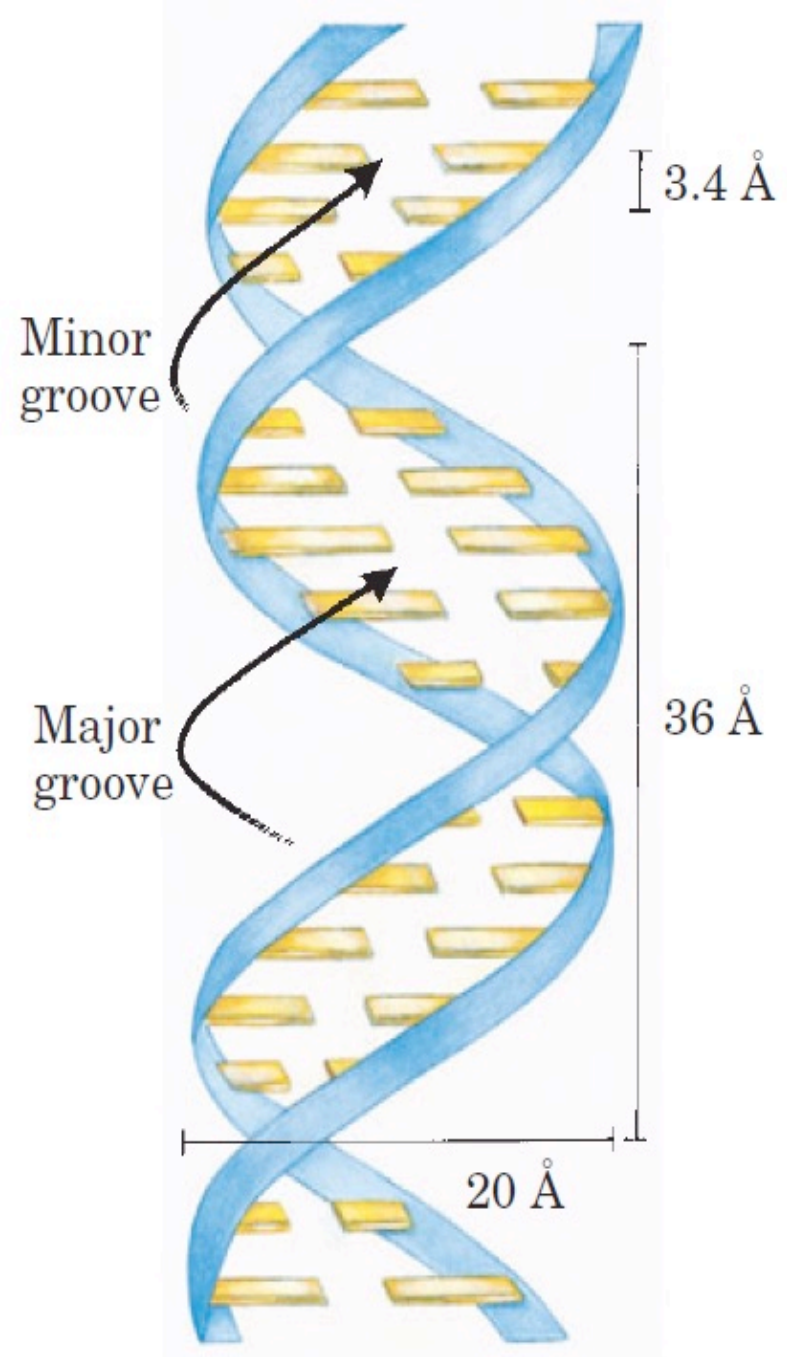
- ❑ Watson, Crick, and Wilkins to propose in the early 1950s a model of a double-stranded DNA molecule.
- ❑ The two strands of this double-stranded helix are held in register by both **hydrogen bonds** between the purine and pyrimidine bases of the respective linear molecules and by **van der Waals** and **hydrophobic interactions** between the stacked adjacent base pairs. The pairings between the purine and pyrimidine nucleotides on the opposite strands are very specific and are dependent upon hydrogen bonding of **A with T** and **G with C**



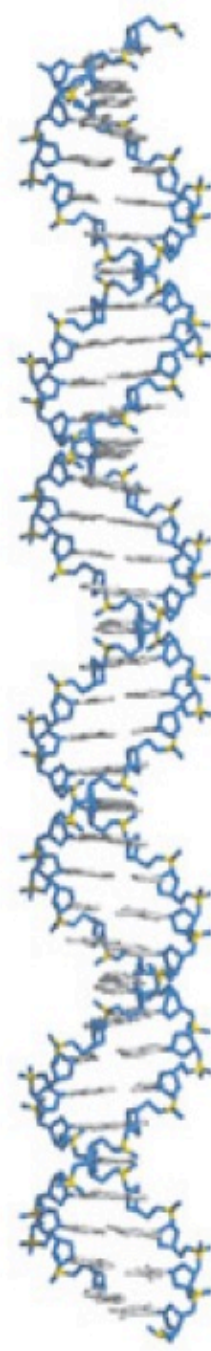




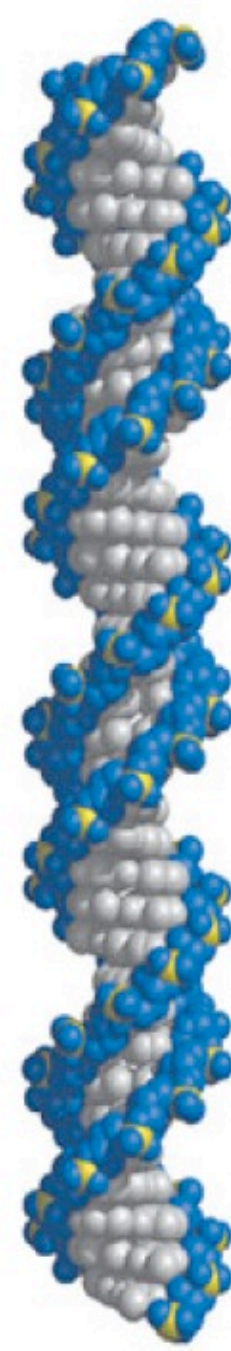




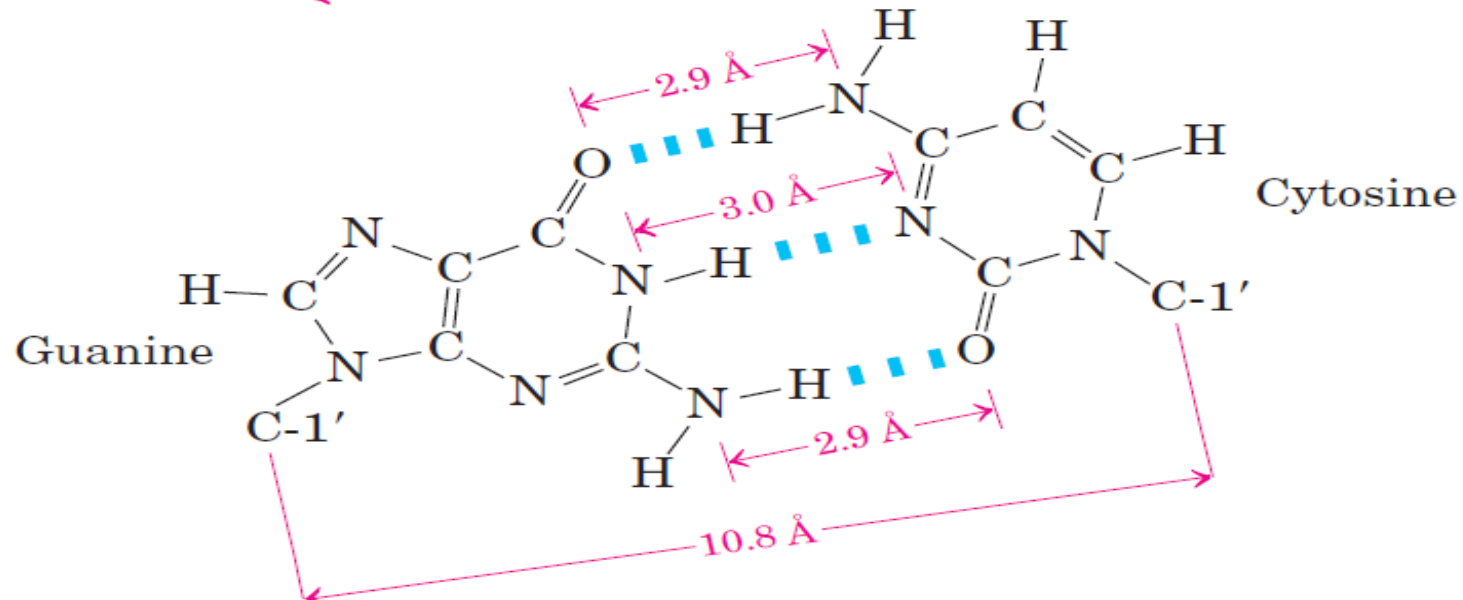
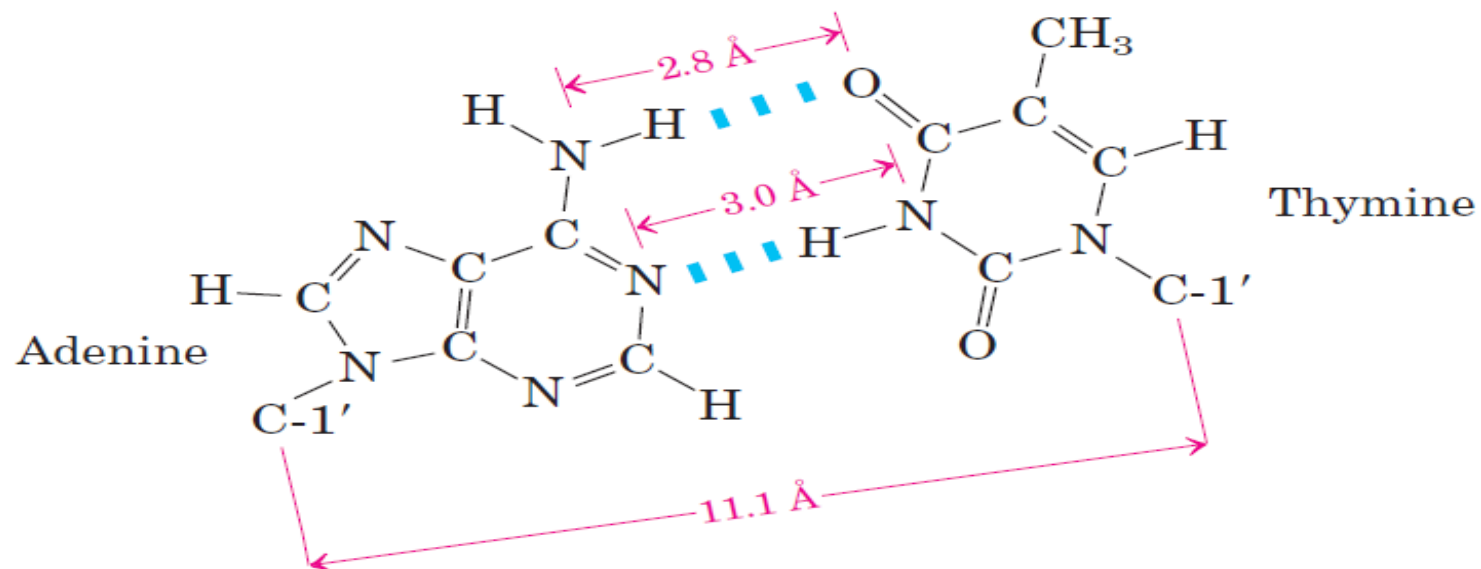
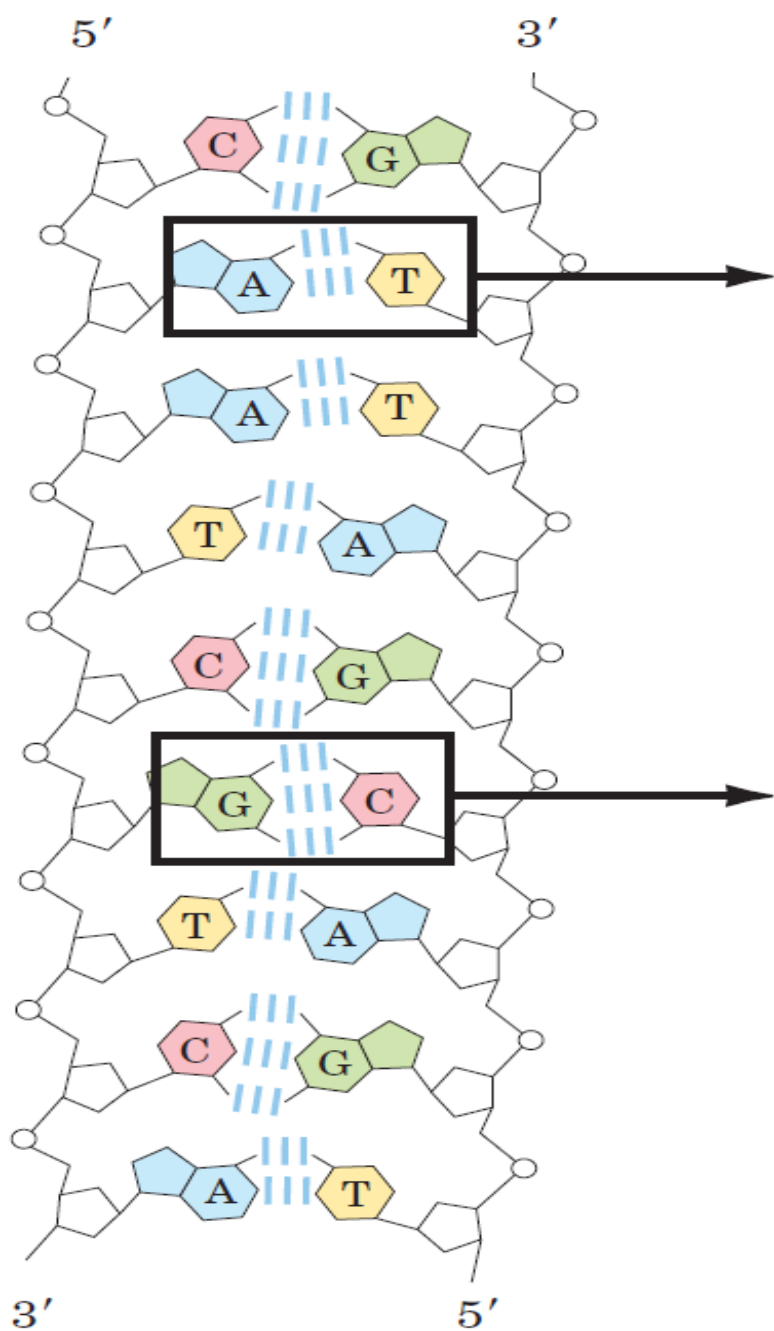
(a)



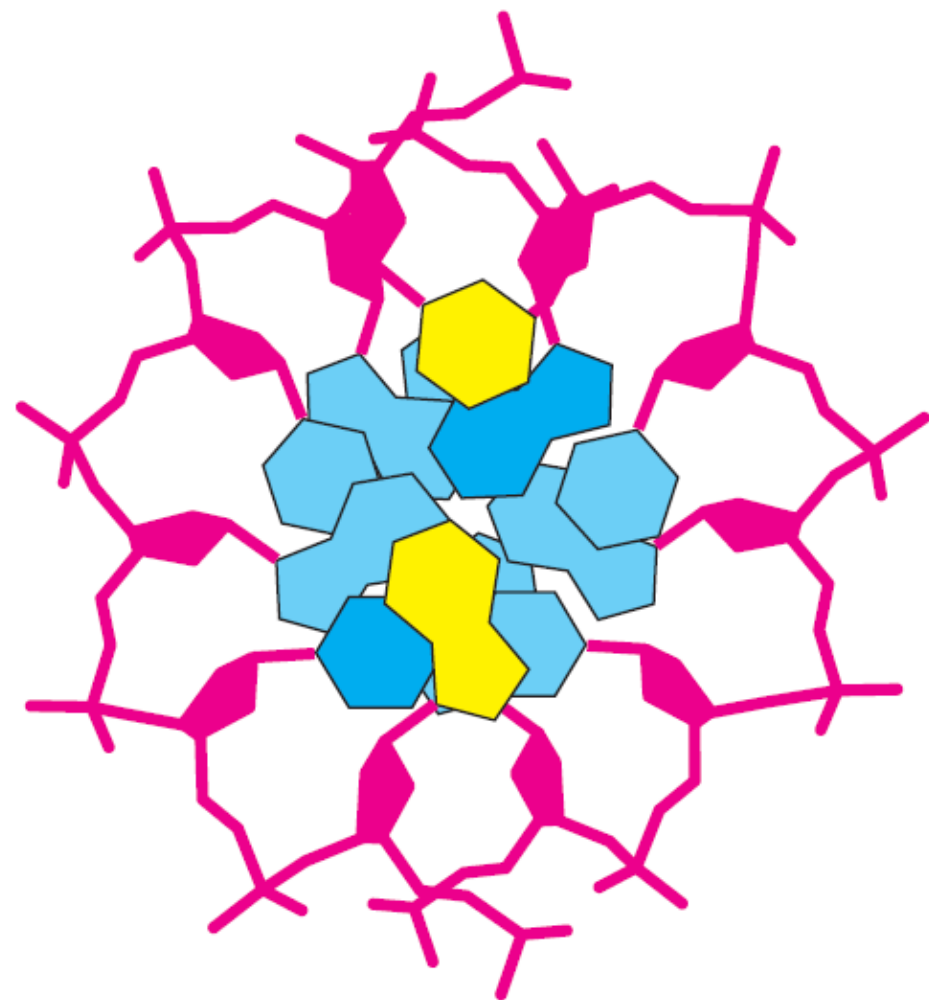
(b)



(c)

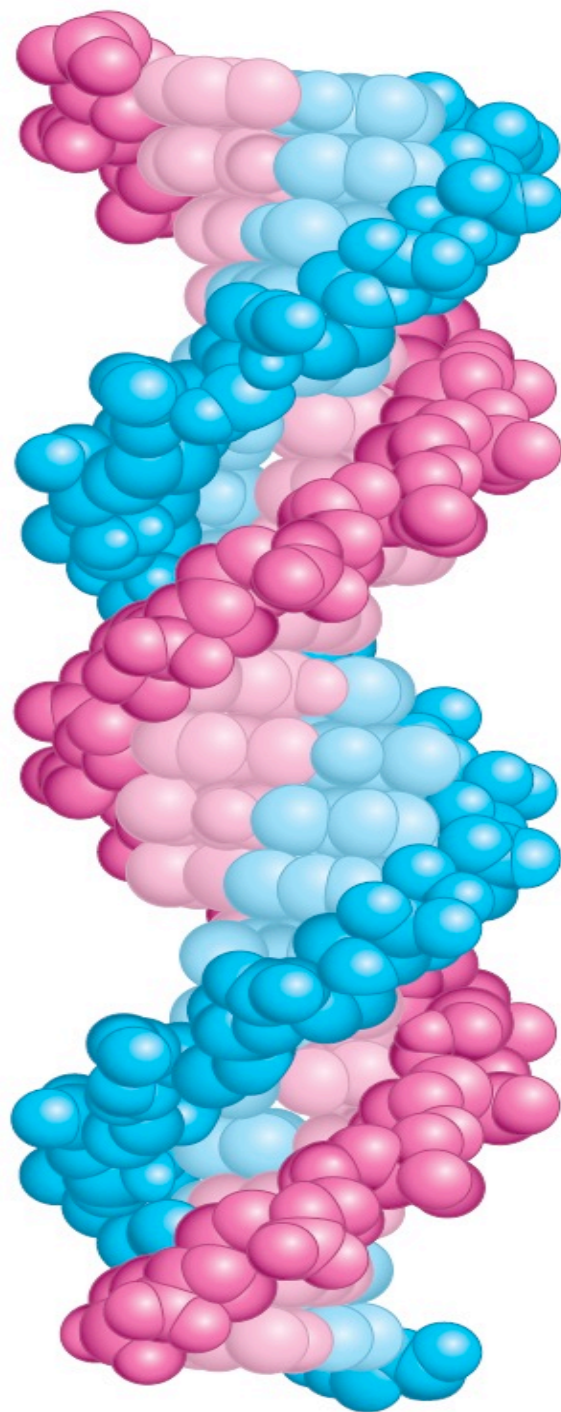


DNA structure

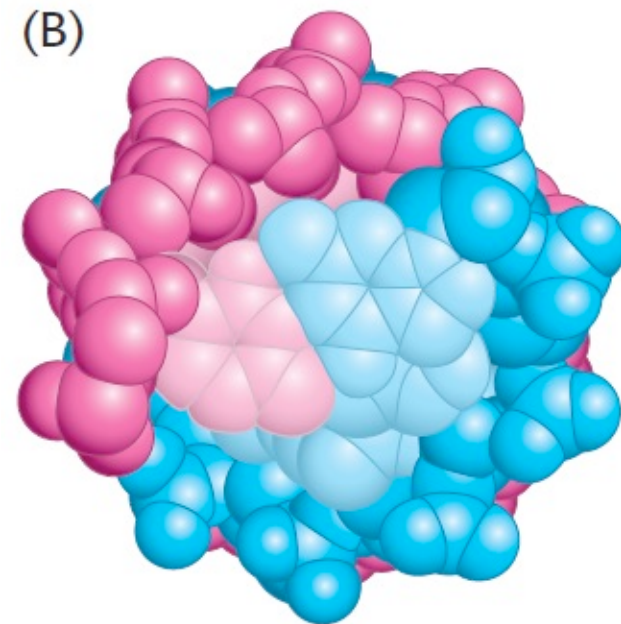


(A)

34Å



(B)



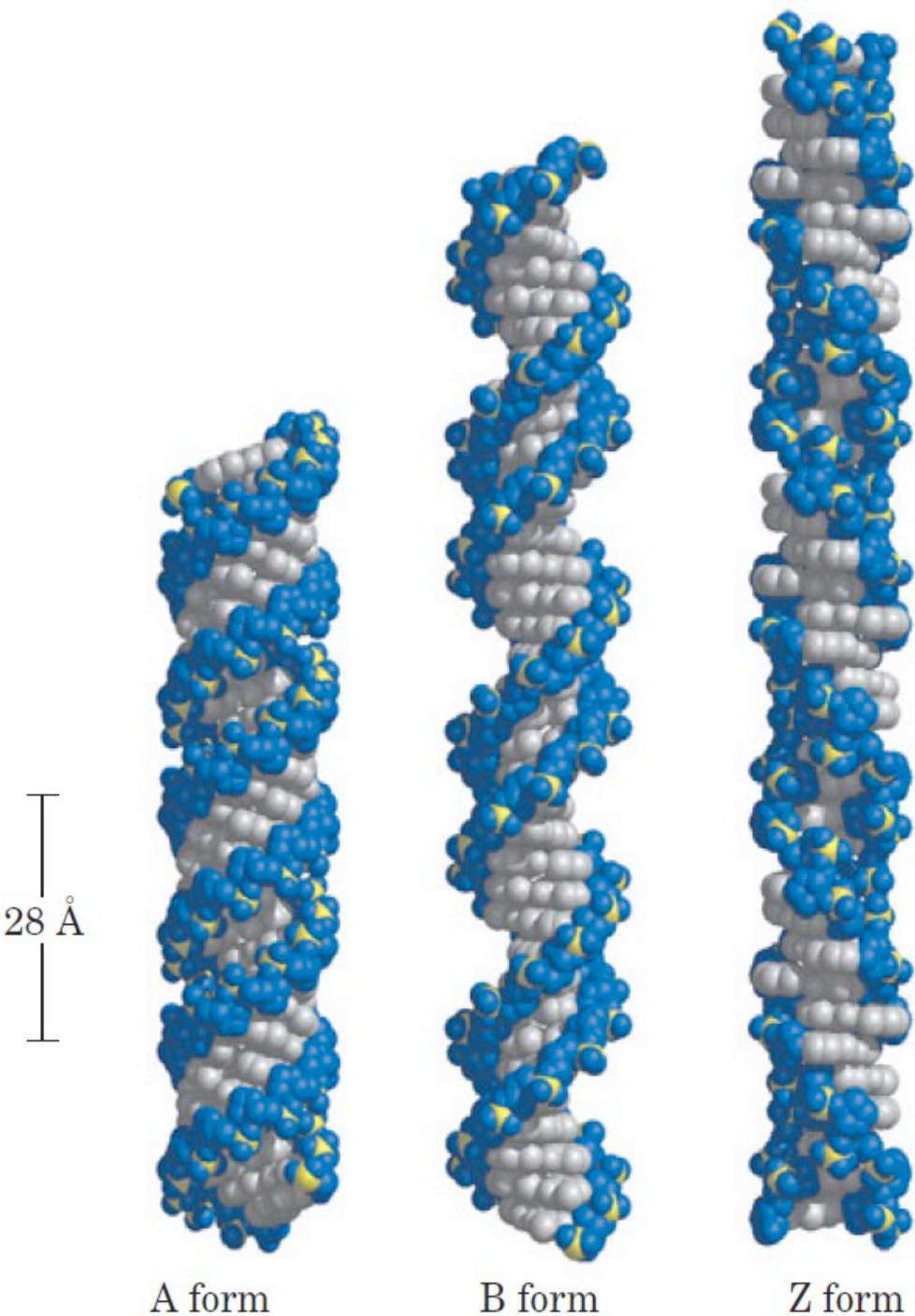
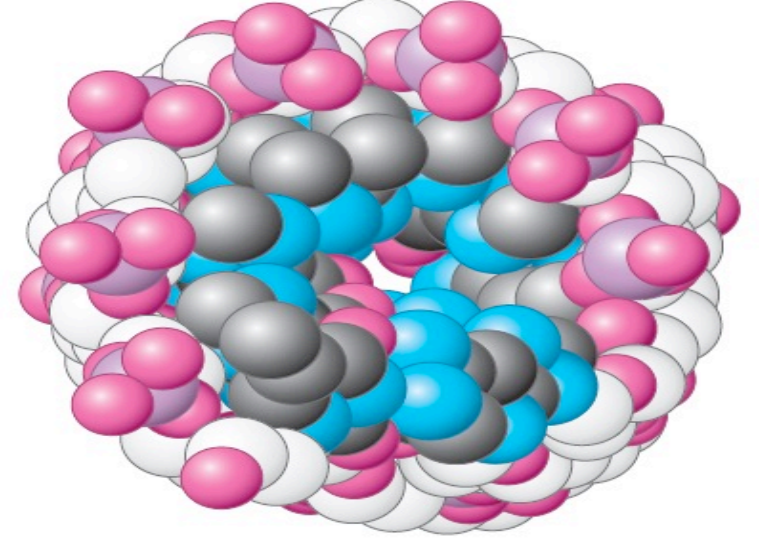
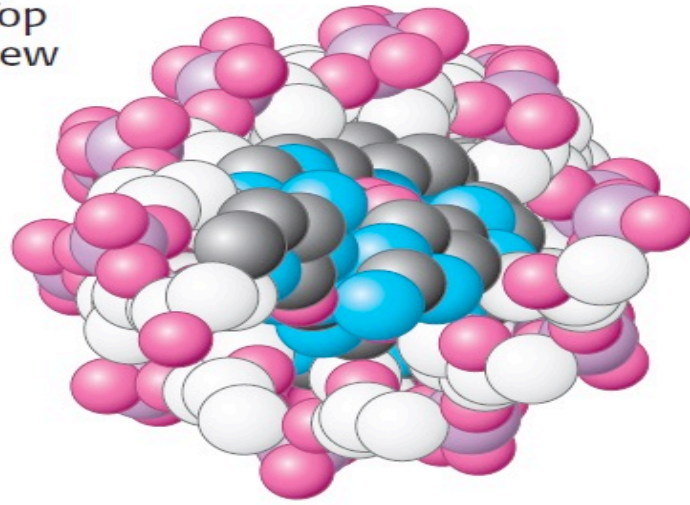


FIGURE 8–17 Comparison of A, B, and Z forms of DNA. Each structure shown here has 36 base pairs. The bases are shown in gray, the phosphate atoms in yellow, and the riboses and phosphate oxygens in blue. Blue is the color used to represent DNA strands in later chapters. The table summarizes some properties of the three forms of DNA.

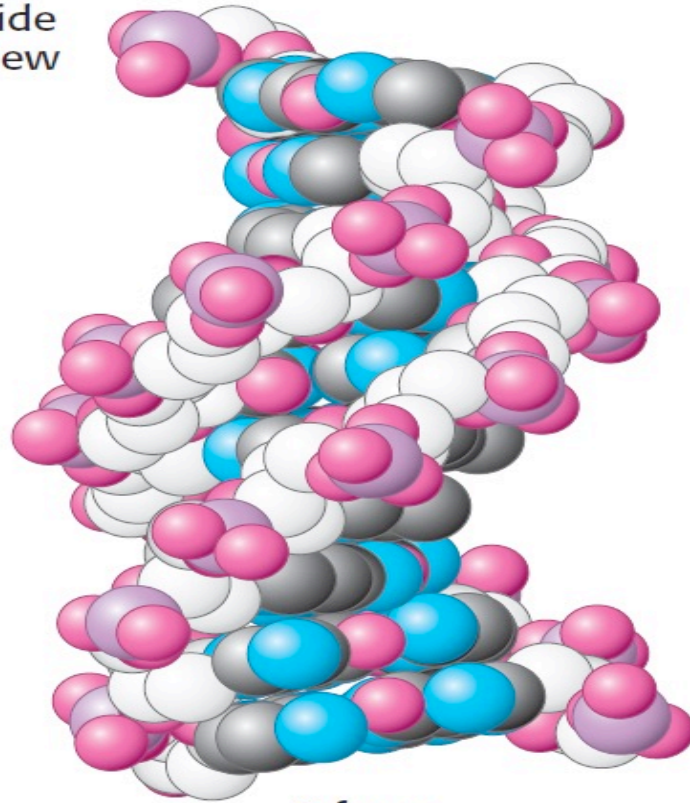
	A form	B form	Z form
Helical sense	Right handed	Right handed	Left handed
Diameter	~26 Å	~20 Å	~18 Å
Base pairs per helical turn	11	10.5	12
Helix rise per base pair	2.6 Å	3.4 Å	3.7 Å
Base tilt normal to the helix axis	20°	6°	7°
Sugar pucker conformation	C-3' endo	C-2' endo	C-2' endo for pyrimidines; C-3' endo for purines
Glycosyl bond conformation	Anti	Anti	Anti for pyrimidines; syn for purines

DNA structure

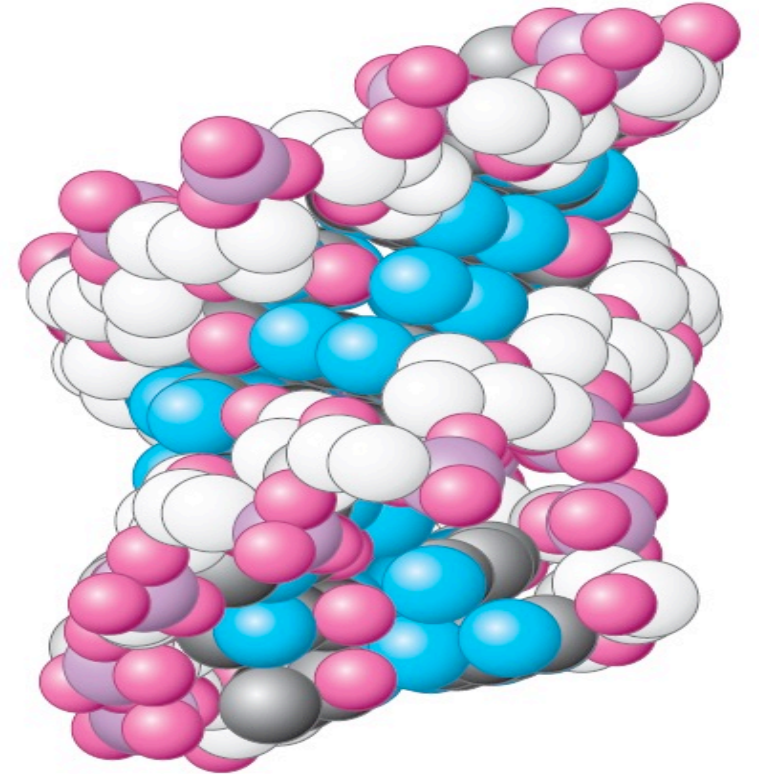
Top
view



Side
view



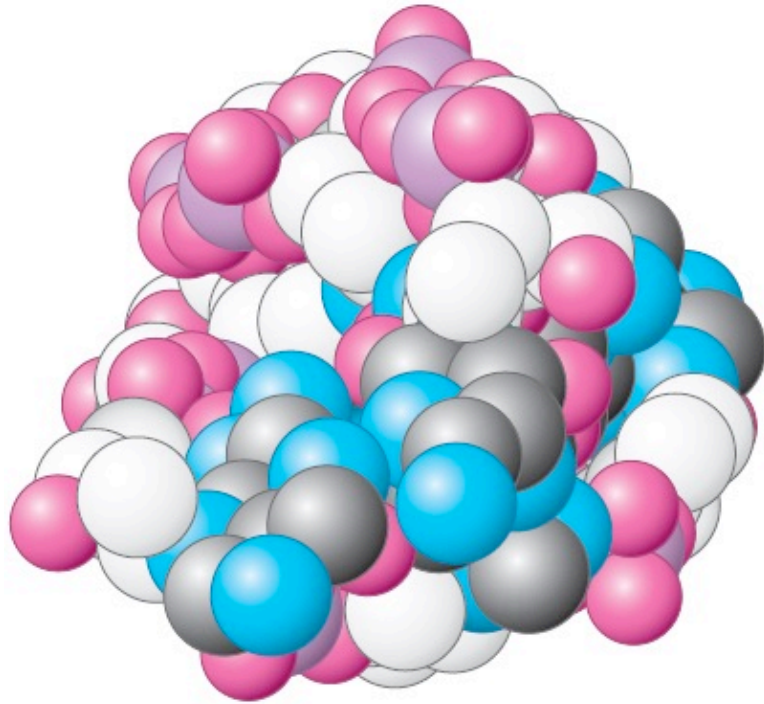
B form



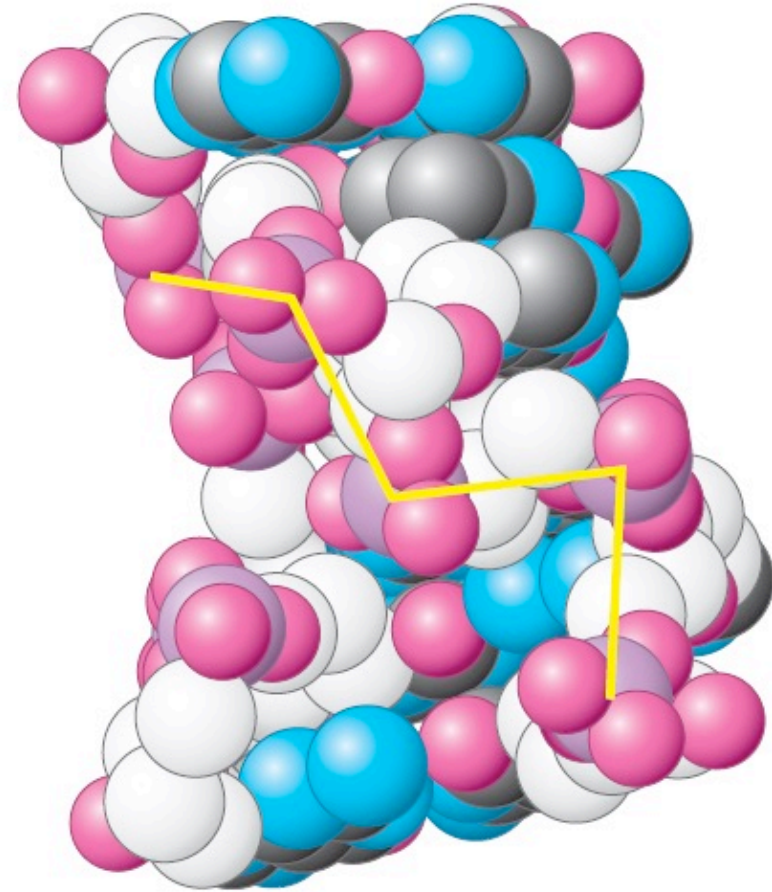
A form



DNA structure Z form



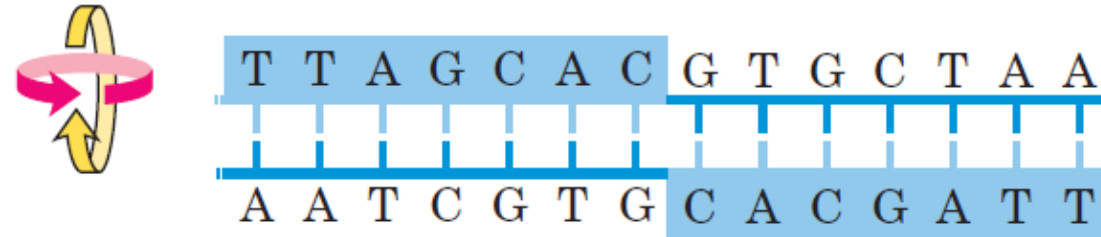
Top view



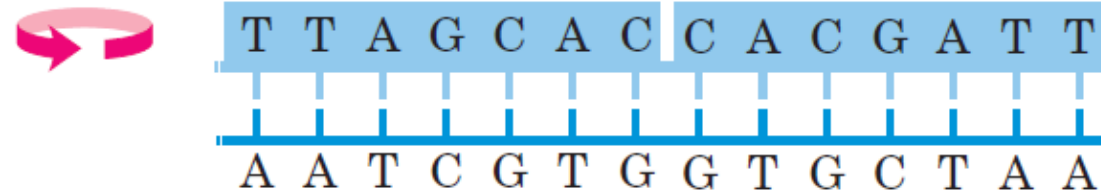
Side view

DNA structure

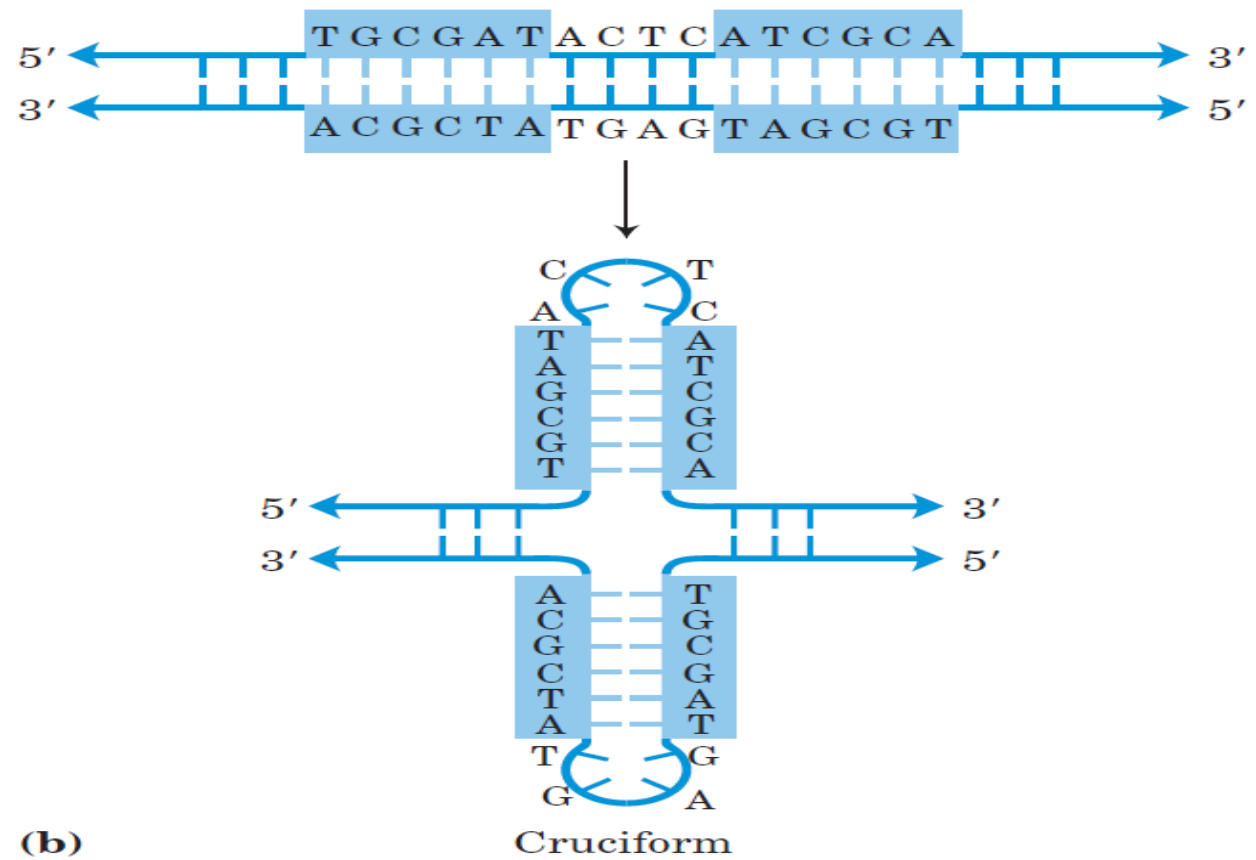
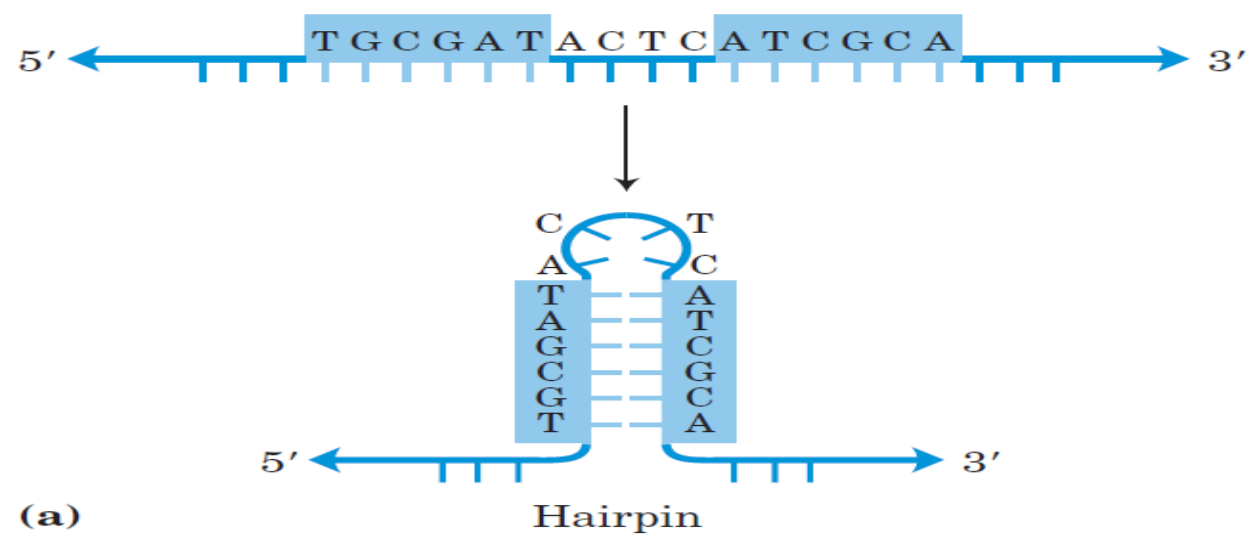
Palindrome



Mirror repeat



DNA structure

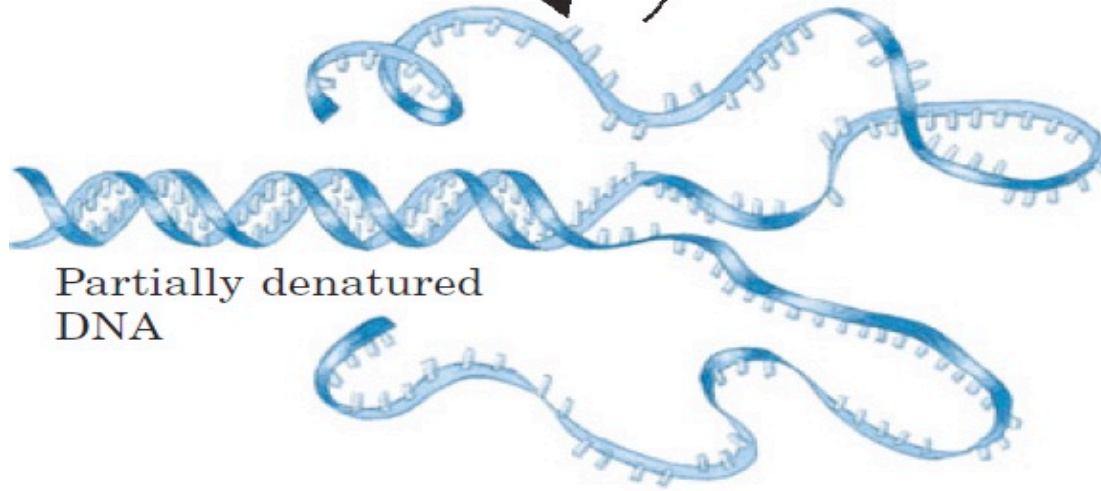




Double-helical
DNA

Denaturation

Annealing



Partially denatured
DNA

Separation
of strands

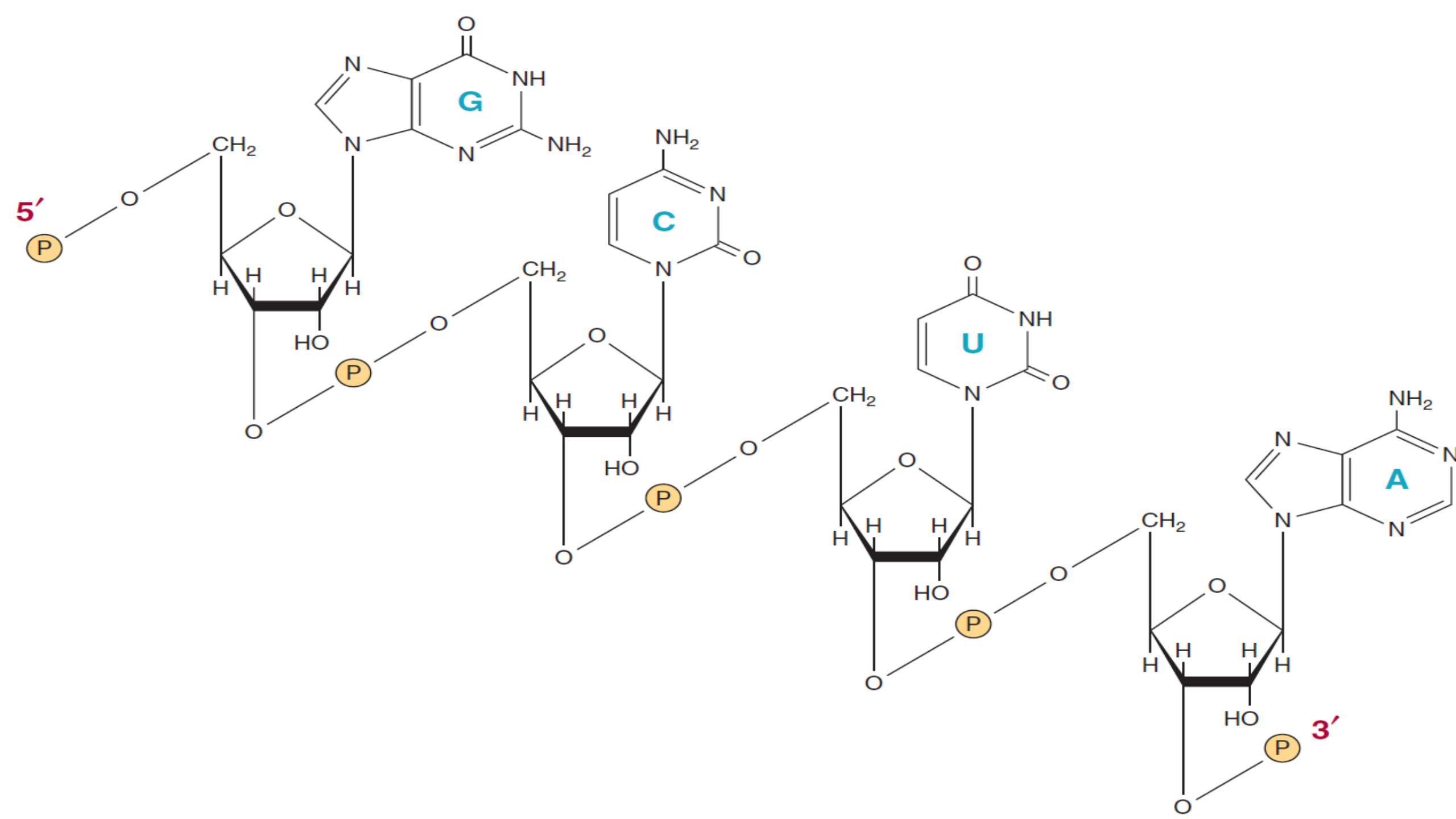
Association of
strands by base
pairing



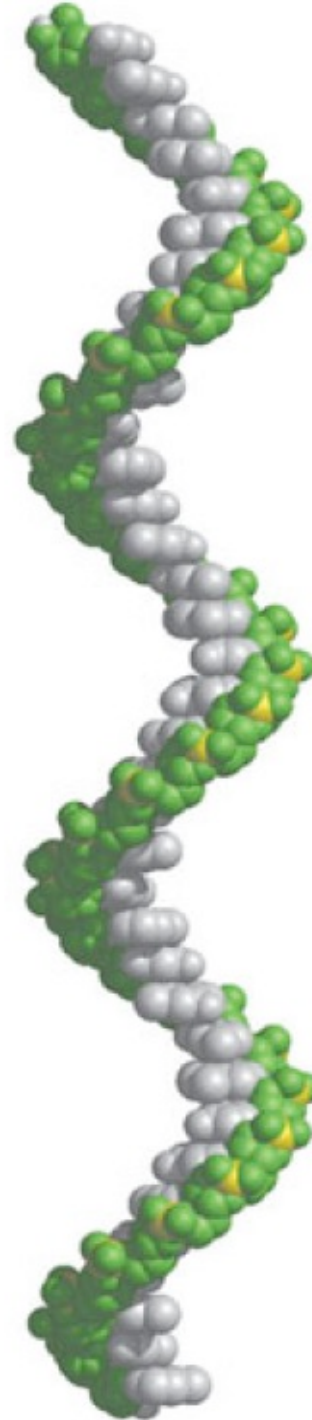
Separated strands
of DNA in random coils

DNA VS RNA

- 1.** In RNA, the sugar moiety is ribose rather than the 2'-deoxyribose of DNA
- 2.** The pyrimidine components of RNA differ from those of DNA.
- 3.** RNA typically exists as a single strand, whereas DNA exists as a double-stranded helical molecule.
- 4.** In RNA G content does not necessarily equal its C content, nor does its A content necessarily equal its U content.
- 5.** RNA can be hydrolyzed by alkali to 2', 3' cyclic diesters of the mononucleotides



RNA structure



[

ima

RNA

- ❑ RNAs have a broader range of functions, and several classes are found in cells.
- ❑ In gene expression, RNA acts as an intermediary by using the information encoded in DNA to specify the amino acid sequence of a functional protein
- ❑ RNA molecules that serve as templates for protein synthesis (ie, that transfer genetic information from DNA to the protein-synthesizing machinery) are designated **mRNAs**.

RNA

- Many other very abundant cytoplasmic RNA molecules (**ribosomal RNAs; rRNAs**) have structural roles wherein they contribute to the formation and function of ribosomes (the organellar machinery for protein synthesis) or serve as adapter molecules (**transfer RNAs; tRNAs**) for the translation of RNA information into specific sequences of polymerized amino acids.

RNA

- ❑ Interestingly, some RNA molecules have intrinsic catalytic activity. The activity of these RNA enzymes, or **ribozymes**, often involves the cleavage of a nucleic acid. Two ribozymes are the peptidyl transferase that catalyzes peptide bond formation on the ribosome, and ribozymes involved in the RNA splicing.
- ❑ In all eukaryotic cells, there are **small nuclear RNA (snRNA)** species that are not directly involved in protein synthesis but play pivotal roles in RNA processing, particularly mRNA processing.

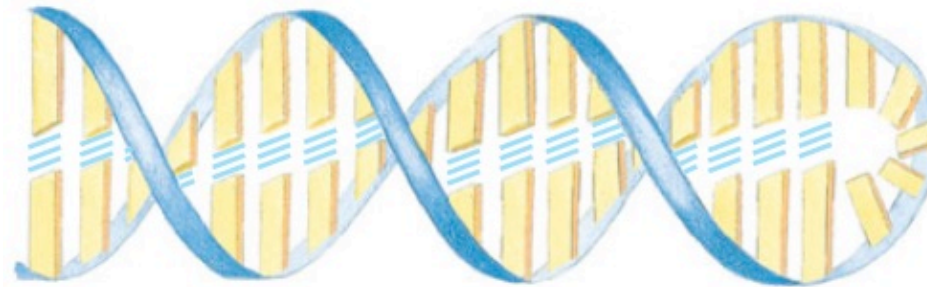
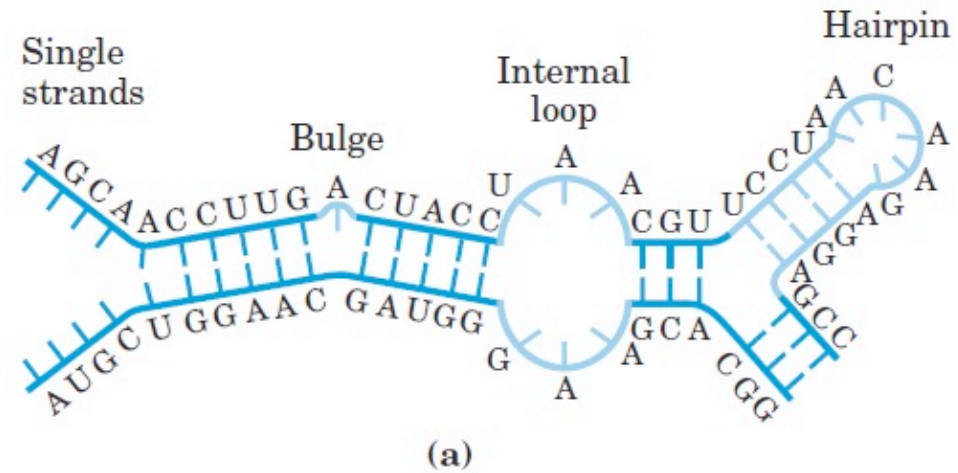
RNA

- ❑ **Ribosomal RNAs (rRNAs)** are components of ribosomes, the complexes that carry out the synthesis of proteins.
- ❑ **Messenger RNAs (mRNAs)** are intermediaries, carrying genetic information from one or a few genes to a ribosome, where the corresponding proteins can be synthesized.
- ❑ **Transfer RNAs (tRNAs)** are adapter molecules that faithfully translate the information in mRNA into a specific sequence of amino acids, covalently linked to an amino acid at one end, they pair with the mRNA in such a way that amino acids are joined to a growing polypeptide in the correct sequence.

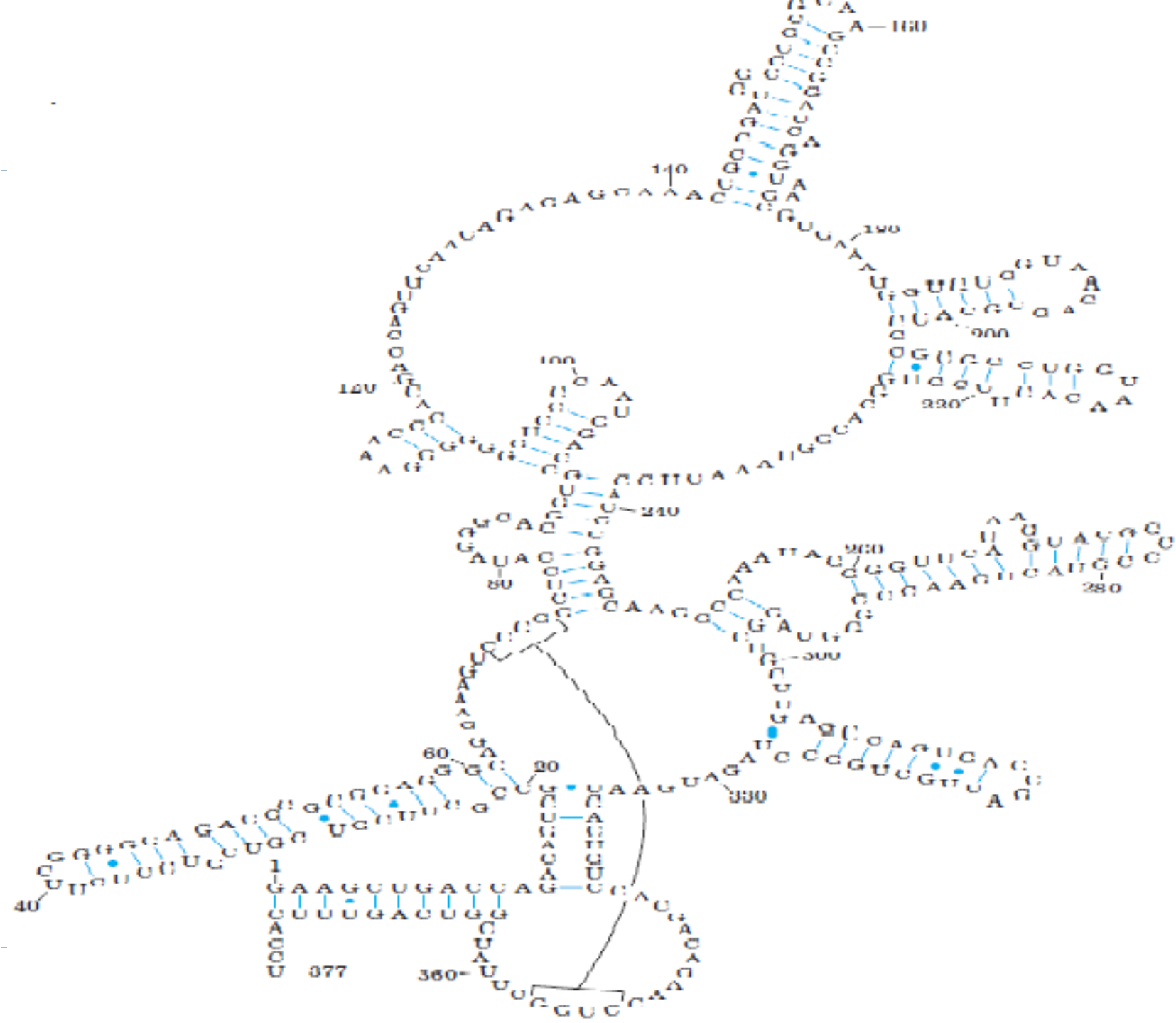
RNA

- ❑ The genetic material for some viruses is RNA rather than DNA
- ❑ Many animal RNA viruses—specifically, the retroviruses (the human immunodeficiency, or HIV virus)—are transcribed by **viral RNA-dependent DNA polymerase, the so-called reverse transcriptase**, to produce a double-stranded DNA copy of their RNA genome.
- ❑ In many cases, the resulting double-stranded DNA transcript is integrated into the host genome and subsequently serves as a template for gene expression and from which new viral RNA genomes and viral mRNAs can be transcribed.

RNA structure



Hairpin double helix
(b)



Messenger RNA

- ❑ the most heterogeneous in abundance, size and stability
- ❑ represents 2% to 5% of total eukaryotic cellular RNA.
- ❑ Although mRNAs from different genes can vary greatly in length, the mRNAs from a particular gene generally have a defined size.
- ❑ The process of forming mRNA on a DNA template is known as **transcription**
- ❑ Eukaryotic mRNAs 5' terminal is “capped” by a 7-methylguanosine triphosphate. The cap is involved in the recognition of mRNA by the translation machinery, and also helps stabilize the mRNA by preventing the nucleolytic attack by 5'-exonucleases

Messenger RNA

- The other end of mRNA molecules, the 3'-hydroxyl terminal, has an attached, nongenetically encoded polymer of A residues 20 to 250 nucleotides in length. The poly(A) “tail” at the 3'-hydroxyl terminal of mRNAs maintains the intracellular stability of the specific mRNA by preventing the attack of 3'-exonucleases and also facilitates translation mRNA

mRNA



(a) Monocistronic

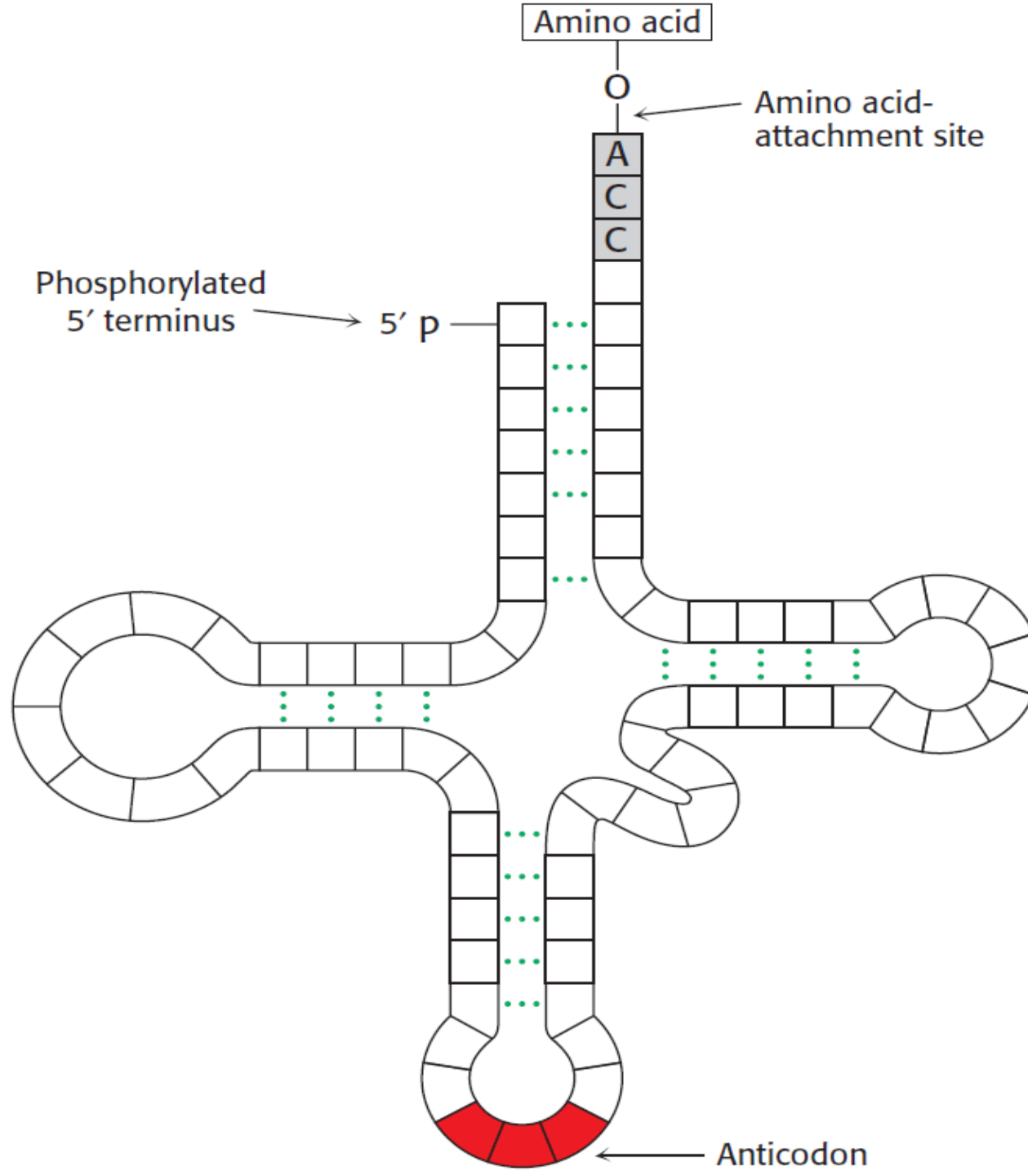


(b) Polycistronic

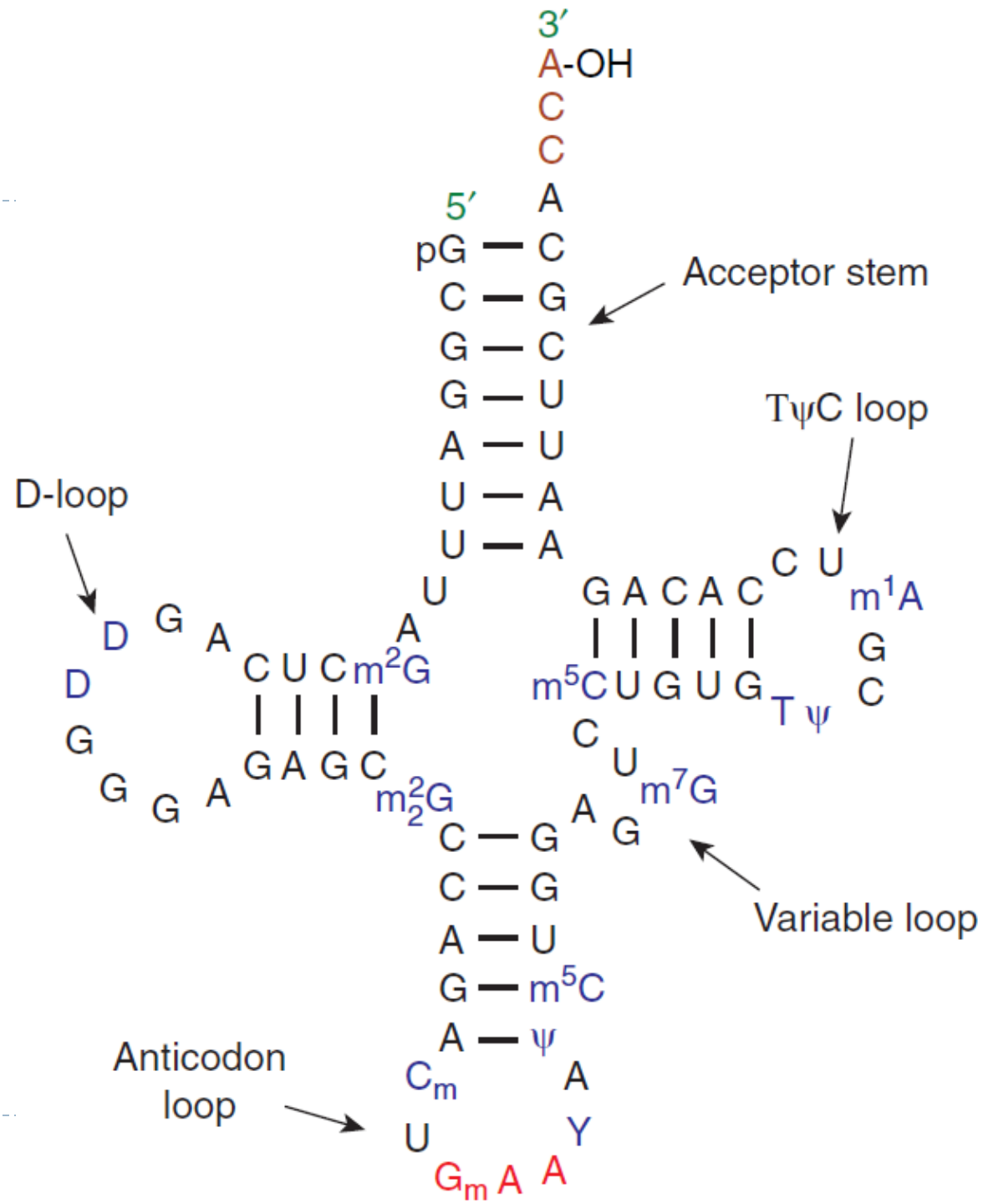
Transfer RNA

- ❑ serve as adapters for the translation of the information in the sequence of nucleotides of the mRNA into specific amino acids.
- ❑ There are at least 20 species of tRNA molecules in every cell, at least one (and often several) corresponding to each of the 20 amino acids required for protein synthesis

tRNA



tRNA



Ribosomal RNA

- ❑ A ribosome is a cytoplasmic nucleoprotein structure that acts as the machinery for the synthesis of proteins from the mRNA templates
- ❑ many ribosomes can be associated with any mRNA molecule to form an assembly called the **Polysome**
- ❑ Represent roughly 70% of total cellular RNA.

Small RNA

- ▣ Most of these molecules are complexed with proteins to form ribonucleoproteins and are distributed in the nucleus, the cytoplasm, or both
- ▣ ***Small Nuclear RNAs*** snRNAs, a subset of the small RNAs, are significantly involved in rRNA and mRNA processing and gene regulation

Large & Small Noncoding Regulatory RNAs

- ▣ NcRNAs exist in two general size classes, large (50-1000nt) and small (20-22nt)
- ▣ ***Micro-RNAs (miRNAs), Silencing RNAs (siRNAs), and Long Noncoding RNAs (lncRNAs)***

ncRNA

- ❑ The **small ncRNAs termed miRNAs and siRNAs typically inhibit of gene expression** at the level of specific protein production by targeting mRNAs through one of several distinct mechanisms
- ❑ siRNAs are generated by the specific nucleolytic processing of large dsRNAs that are either produced from other endogenous RNAs, or dsRNAs introduced into the cell
- ❑ Both **siRNAs and miRNAs typically hybridize, via the formation of RNA–RNA hybridization to their targeted mRNAs**

long noncoding RNAs lncRNAs

- ❑ Do not code for protein, and range in size from ~300 to 1000s of nucleotides in length.
- ❑ These RNAs are typically transcribed from the large regions of eukaryotic genomes that do not encode for protein.
- ❑ In fact transcriptome analyses indicate that **>90% of all eukaryotic genomic DNA is transcribed**. ncRNAs make up a significant portion of this transcription.
- ❑ ncRNAs play many roles ranging from contributing to structural aspects of chromatin to regulation of mRNA gene transcription.